

Approximation Paving over Arbitrary Maximal Abelian Subalgebras

OpenAI

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Abstract

We prove the approximation-paving conjecture of Popa and Vaes for every maximal abelian subalgebra of a complex von Neumann algebra. For each $0 < \varepsilon < 1$, every self-adjoint operator is a strong limit of self-adjoint operators of norm at most three times its norm, each admitting a norm paving with error at most ε times the approximant's own norm. The number of projections is at most $C\varepsilon^{-6}$ for a universal constant C . No separability or conditional-expectation hypothesis is required.

1 Introduction

Let A be a maximal abelian unital $*$ -subalgebra, or MASA, of a complex von Neumann algebra M . For a finite partition $\mathcal{P} = (p_i)_{i=1}^r$ of the unit by projections in A , write

$$C_{\mathcal{P}}(z) = \sum_{i=1}^r p_i z p_i.$$

Norm paving asks whether $C_{\mathcal{P}}(z)$ can be made close in operator norm to an element of A . Approximation paving allows a bounded perturbation of z within any prescribed strong-operator neighborhood before choosing the partition.

Kadison and Singer posed the pure-state extension problem for the atomic diagonal MASA of $\mathcal{B}(\ell^2)$ [KS59]. Anderson formulated its diagonal paving counterpart [Anderson79], and Marcus, Spielman and Srivastava resolved it by mixed characteristic polynomials [MSS15]. For general MASAs, ordinary norm paving of the original operator is too restrictive: in the separable-predual setting, it characterizes type I algebras with a normal expectation onto the MASA [PV15, Theorem 3.3]. Popa and Vaes introduced strong-operator and approximation versions of paving and conjectured that every MASA has both properties [PV15, Definitions 2.2–2.3 and Conjecture 2.8(1)]. We resolve the approximation-paving assertion of that conjecture positively.

Popa proved norm paving in ultraproducts of singular finite MASAs, and hence strong-operator paving in the original finite inclusions [Popa14]. Popa and Vaes established strong-operator and approximation paving for type I algebras with separable predual and for Cartan inclusions in amenable algebras or arising from free ergodic profinite actions [PV15, Sections 3–4]. For singular finite MASAs they obtained quadratic paving bounds [PV15, Theorem 5.1], later sharpened in [PV16]. These restricted cases do not provide the expectation-free approximation theorem below.

Theorem 1.1. *There is a universal constant $C_{\text{ap}} > 0$ such that for every $0 < \varepsilon < 1$ there is an integer $R(\varepsilon) \leq C_{\text{ap}} \varepsilon^{-6}$ with the following property. Let $M \subseteq \mathcal{B}(H)$ be a complex von Neumann algebra, let $A \subseteq M$ be a MASA, and let $x = x^* \in M$. For every finite $F \subset H$ and every $\delta > 0$, there exist $y = y^* \in M$, a partition $\mathcal{P} = (p_i)_{i=1}^{R(\varepsilon)}$ of the unit in A , and $a = a^* \in A$, such that*

$$\|y\| \leq 3 \|x\|, \quad \|(y - x)\xi\| < \delta \quad (\xi \in F),$$

and

$$\|a\| \leq \|y\|, \quad \|C_{\mathcal{P}}(y) - a\| \leq \varepsilon \|y\|.$$

Zero projections may be included in the partition.

Thus the norm-control constant is universal, and the number of projections is independent of the algebra, its representation, the operator, and the strong neighborhood. We do not optimize the sixth-power bound. The relative error in the theorem is measured against the actual approximant's norm; a norming test vector and a final clipping argument will ensure that normalization.

There is also a direct consequence for strong-operator paving when a normal expectation exists. In its definition, the compression is estimated on a large projection whose complement lies in a prescribed strong neighborhood of zero.

Corollary 1.2. *Suppose that a MASA $A \subseteq M$ is the range of a normal conditional expectation. For each $0 < \varepsilon < 1$ there is an integer $R_s(\varepsilon)$, independent of the inclusion, such that for every $x = x^* \in M$ and every strong-operator neighborhood $\mathcal{V}$ of zero, there exist an $R_s(\varepsilon)$ -partition $\mathcal{P}$ in A , an $a = a^* \in A$ with $\|a\| \leq \|x\|$, and a projection $q \in M$ with $\mathbf{1} - q \in \mathcal{V}$, satisfying*

$$\|q(C_{\mathcal{P}}(x) - a)q\| \leq \varepsilon \|x\|.$$

The proof is given in Section 7.4.

The separate companion [OpenAI26, Theorem 1.1] proves a direct quadratic strong-operator paving bound for the original operator. Theorem 1.1 instead norm-paves a nearby bounded operator, with error relative to that approximant's norm. The two main proofs are independent; the conversion in Corollary 1.2 retains its normal-expectation hypothesis. We claim no quadratic bound for $R(\varepsilon)$.

Corollary 1.3 (Quadratic approximation paving for expected MASAs). *There is a universal constant $C > 0$ such that for every $0 < \varepsilon < 1$ one can choose an integer $N(\varepsilon) \leq C\varepsilon^{-2}$ with the following property. If the MASA $A \subseteq M$ is the range of a normal conditional expectation, then the conclusion of Theorem 1.1 holds with $N(\varepsilon)$ in place of $R(\varepsilon)$.*

The proof is given in Section 7.4.

The measured argument. The first part of the proof develops a measurable version of one-sided matrix paving for nonsingular equivalence relations. The finite paving-polynomial framework of [RL20, RS21] supplies the starting point. Its stable expected determinants yield local spectral probabilities and spectral-shift functions. A buffered pinning lemma selects a color while increasing the diagonal penalty by a fixed universal factor and controlling the increase of positive-root mass. The initial root estimate from [RS21, Lemma 24 and proof of Theorem 25] is derived here from [MSS15, Theorems 4.1 and 5.1], using positive-defect completion and deletion of added Gram columns.

Nonsingularity is handled by passing to the Maharam extension, where mass transport is invariant, and weighting the matrix by the height variable. Uniform comparisons between slabs allow the pinning decisions to descend to a single coloring of the base. Sparse random updates then control

the accumulated cost. Only local independence is needed; auxiliary graph colorings do not increase the number of colors in the final paving. The slab comparison and buffered-pinning estimates isolate the two uniformity issues in this passage from finite matrices to measurable operators.

The operator-algebra argument. For a MASA in the centralizer of a faithful normal state, we first pave the algebra generated by its normalizers, using the Feldman–Moore representation [FM77] and the measured theorem. The complementary part has diffuse left–right kernels. We adapt the free-product matrix dilation of [PV15, proof of Theorem 5.1], using random diagonal phases to obtain asymptotic freeness with respect to the induced state. The proof controls the moments and their concentration without assuming traciality, using analytic right multiplication for the modular group.

Finally, a separable reduction preserves the partition bounds. The corner carrying centralizing normal positive functionals has a normal expectation; on the complementary corner, almost disjoint conjugate states produce positive contractions with small diagonal compressions. A three-corner construction preserves the original off-diagonal entries until the final partition is chosen. This is the step that removes the normal-expectation hypothesis.

All new estimates needed for these passages are proved below. We use established structure theorems for conditional expectations, standard form, type I algebras and free products, with their hypotheses specified at the points of application.

Section 2 fixes the state-topology tools; Sections 3–5 prove the measured paving theorem, Section 6 handles the state-preserving case, and Section 7 removes the remaining hypotheses.

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2 State topology and bounded sequence quotients

All algebras and Hilbert spaces are over $\mathbb{C}$. A MASA is automatically a von Neumann algebra: its relative commutant equals itself, by taking real and imaginary parts. Partitions are finite indexed families of orthogonal projections with sum 1; we allow zero projections. For a self-adjoint operator t , let t_+ denote its positive part. We use the same functional-calculus notation for continuous real functions, including

$$\text{clip}_c(t) = \max(-c, \min(t, c)), \quad c > 0.$$

Suppose for this section that M has a faithful normal state φ , and $A \subseteq M_\varphi$ is a MASA in its centralizer. Here

$$M_\varphi = \{a \in M : \varphi(az) = \varphi(za) \text{ for all } z \in M\}.$$

The modular group fixes A pointwise. The conditional-expectation theorem gives a faithful normal φ -preserving expectation $E_A : M \rightarrow A$; see [Takesaki72] or [Hiai20, Theorem 6.6]. Set

$$\|z\|_\varphi = \varphi(z^*z)^{1/2}, \quad \|z\|_\varphi^\# = \|z\|_\varphi + \|z^*\|_\varphi.$$

On bounded sets these seminorms determine, respectively, the σ -strong and σ -strong* topologies. Indeed, in the faithful state representation, $\|z\|_\varphi = \|z\xi_\varphi\|$, and $M'\xi_\varphi$ is dense. Boundedness therefore promotes convergence on ξ_φ to strong convergence on all vectors. The bounded σ -strong topology is intrinsic to the von Neumann algebra. Apply the same reasoning to adjoints for the second assertion. In standard $L^2(M)$ notation we write $\xi_\varphi = \varphi^{1/2}$. Elements of A commute with this vector under the left and right actions.

Lemma 2.1. *Every A -partition compression C_Q preserves φ and is contractive for both state seminorms. As Q refines through all finite A -partitions,*

$$C_Q(z) \longrightarrow E_A(z) \quad \text{in } \|\cdot\|_\varphi^\# \quad (z \in M).$$

The same holds along any refining sequence of partitions generating A .

Proof. The centralizer identity gives $\varphi(C_Q(z)) = \varphi(z)$. Compression is unital completely positive, so its Schwarz inequality gives state-norm contraction; apply it to adjoints as well. On the state Hilbert space the compression is implemented by the orthogonal projection

$$\zeta \longmapsto \sum_{q \in Q} q\zeta q.$$

These projections decrease under refinement. Their action on $z\varphi^{1/2}$ is $C_Q(z)\varphi^{1/2}$. Meanwhile the operators $C_Q(z)$ are uniformly bounded. Every ultraweak cluster point commutes with all projections of A , and hence is in A . Since $E_A C_Q(z) = E_A(z)$, the cluster point equals $E_A(z)$. The Hilbert-space limit of the decreasing projections must therefore be $E_A(z)\varphi^{1/2}$. Repeating for z^* proves the claim. For a generating refining sequence, commuting with all of its partition projections is already equivalent to commuting with A , so the same proof applies. $\square$

The following quotient allows us to record vanishing state errors while retaining norm estimates. Its multiplier condition matters when φ is not tracial.

Lemma 2.2. *Let*

$$\mathcal{I}_\varphi = \{(z_j) \in \ell^\infty(M) : \|z_j\|_\varphi^\# \rightarrow 0\},$$

and let

$$\mathcal{C}_\varphi = \{t \in \ell^\infty(M) : t\mathcal{I}_\varphi \subseteq \mathcal{I}_\varphi, \mathcal{I}_\varphi t \subseteq \mathcal{I}_\varphi\}.$$

Then $\mathcal{C}_\varphi$ is a unital C^ -algebra and $\mathcal{I}_\varphi$ is a closed two-sided ideal in it. Constant sequences from M and all sequences in $\ell^\infty(A)$ belong to $\mathcal{C}_\varphi$.*

Proof. The set $\mathcal{I}_\varphi$ is self-adjoint and norm closed. If $s, t \in \mathcal{I}_\varphi$, then

$$\|s_j t_j\|_\varphi \leq \|s_j\| \|t_j\|_\varphi, \quad \|(s_j t_j)^*\|_\varphi \leq \|t_j\| \|s_j^*\|_\varphi,$$

so $st \in \mathcal{I}_\varphi$. The two multiplier conditions now show directly that $\mathcal{C}_\varphi$ is a unital, norm-closed $*$ -algebra, contains $\mathcal{I}_\varphi$, and makes it an ideal. Multiplication by a fixed bounded operator is continuous on bounded sets for strong $*$ convergence, so constant sequences are multipliers.

If $a \in A$, then left multiplication has state-norm bound $\|a\|$, and the centralizer property gives

$$\|ta\|_\varphi^2 = \varphi(a^* t^* ta) = \varphi(t^* t a a^*) \leq \|a\|^2 \varphi(t^* t).$$

The last inequality follows from positivity, since $\varphi(t^* tb) = \varphi(b^{1/2} t^* t b^{1/2}) \geq 0$ for $b \geq 0$ in A . The same estimates apply to adjoints, uniformly on bounded A -sequences. Thus these sequences are multipliers. $\square$

Write $[t_j]$ for an image in $\mathcal{C}_\varphi/\mathcal{I}_\varphi$. A sequence of A -partitions of uniformly bounded size acts by compression in this quotient. If q refines p , its compression applied to a p -compression is its compression of the original element. Since these maps are unital and positive, a common refinement preserves separate scalar upper and lower bounds for a self-adjoint element. In particular it preserves a norm bound.

We will also use the following direct consequence of functional calculus. If a self-adjoint multiplier sequence T_j has $\|[T_j]\| \leq c$, then

$$w_j = T_j - \text{clip}_c(T_j) \in \mathcal{I}_\varphi, \quad \|T_j - w_j\| \leq c.$$

If T_j is block diagonal for a partition, so is w_j . This converts the quotient estimates below into explicit corrections.

3 Finite mixed polynomials and buffered pinning

We develop a rule for fixing one vertex's color while controlling the total positive-root mass of an expected characteristic polynomial. The rule will be applied repeatedly in the measured paving argument; its constants must therefore be independent of the matrix size and of previous color choices. We first identify the rooted mass and the change caused by one pin, then prove the inequality that relates them.

3.1 Stable polynomials and local spectral data

Let Q be a finite set, let $W = (W_{vw})_{v,w \in Q}$ be Hermitian with zero diagonal, and let $a_v \in \mathbb{R}$. At each v , independently choose a color in $\{1, \dots, r\}$ with probabilities $t_{vi} \geq 0$, $\sum_i t_{vi} = 1$. The matrix W_{pav} retains the entries between vertices of the same color and sets all other entries to zero. Define

$$f_Q(Y) = \mathbb{E} \det(\text{diag}(Y_v + a_v)_{v \in Q} - W_{\text{pav}}), \quad f_\emptyset = 1. \quad (3.1)$$

All restrictions to subsets of Q use the restricted matrix, penalties, and probabilities. The polynomial is real and multiaffine, its leading monomial is $\prod_{v \in Q} Y_v$, and

$$\partial_{Y_v} f_Q = f_{Q \setminus \{v\}}. \quad (3.2)$$

We write $f_Q(z)$ when every variable is set equal to z . The uniform-probability paving polynomial and its determinant-derivative representation occur in [RL20, Theorem 1.6 and Proposition 1.7]; independent nonuniform distributions are also discussed in [RS21, remark following Lemma 24]. We give the identities and stability arguments in the form needed for local perturbations below.

Lemma 3.1 (Stability). *Put $t_i^{1/2} = \text{diag}(\sqrt{t_{vi}})_{v \in Q}$. Then*

$$f_Q(Y) = \left(\prod_{v \in Q} \frac{1}{r!} \partial_{Y_v}^{r-1} \right) \prod_{i=1}^r \det(\text{diag}(Y + a) - r t_i^{1/2} W t_i^{1/2}). \quad (3.3)$$

This identity remains valid for complex t_{vi} with $\sum_i t_{vi} = 1$, where the expectation means the finite algebraic sum with the corresponding product weights, and the same square root is used in both occurrences of $t_i^{1/2}$. The algebraic identity also holds with complex penalties and arbitrary complex off-diagonal entries of W , still with zero diagonal.

More generally, let $\mathcal{H}$ be an open half-plane. If the product of determinants on the right of (3.3), for every principal restriction, is nonzero whenever all its variables lie in $\mathcal{H}$, then the same holds for f_Q . In particular, for Hermitian W and real probabilities and penalties, f_Q is upper-half-plane stable and its univariate specialization is real-rooted.

Proof. Expand each determinant by permutations. Since W has zero diagonal, the off-diagonal terms are collections of disjoint nontrivial permutation cycles. Each such cycle C has sign minus and weight

$$w_C = \left(\prod_{(v,w) \in C} W_{vw} \right) \sum_{i=1}^r \prod_{v \in C} t_{vi} \quad (3.4)$$

in the expectation; unused vertices contribute $Y_v + a_v$. Indeed, all vertices on a cycle must receive one common color, and disjoint cycles involve disjoint independent choices.

In the product on the right of (3.3), suppose a vertex occurs in nontrivial cycles of k different determinant factors. Its degree in $Y_v + a_v$ is $r - k$. If $k \geq 2$, the indicated derivative kills the term. If $k = 0$, the normalized derivative sends $(Y_v + a_v)^r$ to $Y_v + a_v$. If $k = 1$, it sends $(Y_v + a_v)^{r-1}$ to $1/r$, cancelling the factor r from the matrix scaling at that vertex. The remaining terms are precisely (3.4) for disjoint cycles. The two square-root factors at each cycle vertex multiply to t_{vi} , so this argument also proves the complex-parameter identity independently of the choices of square roots. This cycle calculation is algebraic in the entries of W and does not require them to be Hermitian; that hypothesis is used for the real-data stability conclusion, not for the identity.

For the half-plane assertion, proceed by induction on the subset D of vertices at which $r - 1$ derivatives have already been taken, simultaneously for all principal restrictions. Fix the other variables in $\mathcal{H}$ and take a vertex $v \notin D$. In each determinant, the coefficient of Y_v is its principal determinant with v removed, even for complex penalties. Extracting the coefficient of Y_v^r from the product commutes with all normalized derivatives at vertices in D . Thus this coefficient is exactly the already-processed product for the principal set with v removed. It is nonzero by induction, so the polynomial in Y_v has degree exactly r . Its roots lie in the closed convex complement of $\mathcal{H}$. Gauss–Lucas implies that each of its successive derivatives has all roots in that complement and is nonzero. This proves the induction step.

For completeness, the determinant hypothesis holds whenever $\mathcal{H}$ lies strictly beyond a numerical-range bound for

$$B_i = rt_i^{1/2} W t_i^{1/2} - \text{diag}(a). \quad (3.5)$$

For example, if $\text{Im} \langle B_i u, u \rangle \leq \beta \|u\|^2$ and $\text{Im} Y_v > \beta$ for all v , a null vector of $\text{diag}(Y) - B_i$ contradicts the imaginary part of its quadratic form. The same reasoning applies after a common rotation of the half-plane; principal compressions inherit numerical-range bounds. For Hermitian W and real probabilities and penalties, the matrices (3.5) are Hermitian, giving upper-half-plane stability. Real coefficients and complex conjugation then give real-rootedness of the univariate specialization. $\square$

Lemma 3.2 (Local spectral data). *Fix $v \in Q$, put $f = f_Q$, $g = f_{Q \setminus \{v\}}$, and write $h = f/g$, $G = 1/h$ at univariate arguments. For $z \in \mathbb{C}$ with $\text{Im } z > 0$,*

$$\text{Im } h(z) \geq \text{Im } z. \quad (3.6)$$

If s_j are the roots of g , with multiplicities, there are $\nu_j \geq 0$ such that

$$h(z) = z + a_v - \sum_j \frac{\nu_j}{z - s_j}. \quad (3.7)$$

Consequently, on choosing $|c_j|^2 = \nu_j$,

$$B = \begin{pmatrix} -a_v & c^* \\ c & S \end{pmatrix}, \quad S = \text{diag}(s_j), \quad \det(zI - B) = f(z). \quad (3.8)$$

The function G is the transform

$$G(z) = \int_{\mathbb{R}} \frac{d\gamma_v(x)}{z - x} \quad (3.9)$$

of the spectral probability measure of B at its first coordinate. Its first two moments are

$$\begin{aligned} \int x d\gamma_v(x) &= -a_v, & \int x^2 d\gamma_v(x) &= a_v^2 + \sum_j \nu_j, \\ \sum_j \nu_j &= \sum_{w \neq v} |W_{vw}|^2 \sum_i t_{vi} t_{wi}. \end{aligned} \quad (3.10)$$

Let $N_f(x)$ and $N_g(x)$ count roots strictly greater than x , including multiplicities, and define

$$\xi_h(x) = N_f(x) - N_g(x), \quad \ell_v = \int_0^\infty \xi_h(x) dx, \quad m_v = \int_{\mathbb{R}} x_+ d\gamma_v(x). \quad (3.11)$$

Then $\xi_h = \mathbf{1}_{\{h < 0\}}$ off a finite subset of $\mathbb{R}$, and

$$\frac{1}{\pi} \arg h(x + i\eta) = (P_\eta * \xi_h)(x), \quad P_\eta(x) = \frac{\eta}{\pi(x^2 + \eta^2)}, \quad \eta > 0, \quad (3.12)$$

where $\arg h \in (0, \pi)$. Moreover,

$$\sum_{v \in Q} G_{v,Q}(z) = \frac{f'_Q(z)}{f_Q(z)}, \quad \sum_{v \in Q} \gamma_v = \sum_{f_Q(\lambda)=0} \delta_\lambda, \quad (3.13)$$

with multiplicities on the right.

If v is made deterministic of color i , write h_i for the new quotient relative to the same g , and put $R_i = h_i/h$. Its representation has the same a_v and poles s_j , and masses $\nu_{ij} \geq 0$ which can be chosen so that

$$h = \sum_i t_{vi} h_i, \quad \nu_j = \sum_i t_{vi} \nu_{ij}. \quad (3.14)$$

If the penalty at v is also changed to Ka_v , the quotient becomes

$$h'_i = h_i + (K - 1)a_v. \quad (3.15)$$

Here the prime with a color index denotes the buffered quotient, not a derivative. We write $\ell'_{v,i}$ for the positive-axis integral of the spectral shift defined using h'_i and the unchanged g .

Proof. Fix every variable except Y_v at z in the upper half-plane. By (3.2), division by $g(z) \neq 0$ produces a monic affine polynomial $Y_v - \alpha(z)$. Its root cannot be in the upper half-plane by Lemma 3.1. Hence $h(z) = z - \alpha(z)$ satisfies (3.6).

The rational function h has real coefficients and only real poles. Its upper-half-plane property rules out poles of order two or higher: the imaginary part of a real leading polar term of such order has both signs along rays approaching the pole from above. At an actual simple pole the residue must be negative, as seen by approaching vertically. The polynomial part is $z + a_v$, by comparing the two highest coefficients of the monic polynomials f and g . This gives (3.7). For repeated roots of g , assign the total residue to one copy and pad with zero masses at the others. The Schur complement proves (3.8) and (3.9), first away from the poles and then as identities of rational functions. The first two moments follow from the first diagonal entries of B and B^2 . Expanding f/g at infinity, the coefficient of $1/z$ is minus the sum of the length-two cycle weights through v , giving the final identity in (3.10).

For real x away from roots and poles, the Schur complement of $xI - S$ in $xI - B$ is $h(x)$. Inertia is additive under this congruence, so the number of negative eigenvalues of $xI - B$ exceeds that of $xI - S$ by exactly $\mathbf{1}_{\{h(x) < 0\}}$. This proves the assertion about ξ_h and in particular $0 \leq \xi_h \leq 1$ almost everywhere.

Factor f and g over their real roots. On the upper half-plane, take each factor $z - \lambda$ with argument in $(0, \pi)$. The sum of the numerator arguments minus the denominator arguments equals $\arg h$: their difference is a continuous multiple of 2π and tends to zero at $z = iR$ as $R \rightarrow \infty$. The argument of $x + i\eta - \lambda$, divided by π , is the Poisson integral of $\mathbf{1}_{(-\infty, \lambda)}$. Summing and subtracting gives (3.12). Finally the chain rule and (3.2) give the first identity in (3.13). The second follows by comparing the finite Stieltjes transforms on the two sides.

Pinning changes only the probability at v , so g is unchanged and f is the weighted sum of its pinned versions. This gives the first identity in (3.14). Each pinned polynomial is stable by Lemma 3.1, and thus has the same type of quotient representation. Comparing residues at each distinct pole gives the second identity; use the same placement on repeated copies for every choice. Finally changing the v -penalty adds $(K - 1)a_v g$ to the polynomial, proving (3.15). $\square$

The two local quantities have different roles. Define the total positive-root mass, with roots counted according to multiplicity, by

$$\mathcal{E}(Q) = \sum_{f_Q(\lambda)=0} \lambda_+. \quad (3.16)$$

The sum-of-measures identity and the definition of the spectral shift give

$$\mathcal{E}(Q) = \sum_{v \in Q} m_v, \quad \ell_v = \mathcal{E}(Q) - \mathcal{E}(Q \setminus \{v\}). \quad (3.17)$$

Thus m_v distributes the total cost among the vertices, whereas ℓ_v is the cost of inserting v . A buffered pin at v leaves the deleted polynomial unchanged, so its change of total cost is exactly $\ell'_{v,i} - \ell_v$. These are costs of the expected characteristic polynomial; in general they are not expectations of the positive-part trace of the randomly paved matrix. They become the ordinary spectral costs when all colors are deterministic.

3.2 The initial largest-root bound

The following estimate is obtained in [RS21, Lemma 24 and proof of Theorem 25]. We include a derivation from the MSS covariance bound, explaining both the positive-defect completion and the deletion of the added Gram columns.

Lemma 3.3 (Initial largest-root bound). *Let T be a Hermitian contraction on a nonempty finite index set Q , with zero diagonal. In (3.1), take $W = T$, $a_v = 0$, and $t_{vi} = 1/r$. Then*

$$\max \text{root}(f_Q) \leq 2\sqrt{2/r} + 2/r. \quad (3.18)$$

In particular, giving every vertex the same penalty b places all roots strictly below zero if $2\sqrt{2/r} + 2/r < b$.

Proof. Set $D_0 = I + T$. Then $0 \leq D_0 \leq 2I$ and $(D_0)_{vv} = 1$. Independently form the columns

$$u_v = D_0^{1/2} e_v \otimes e_{\text{color}(v)} \quad \text{in } \mathbb{C}^Q \otimes \mathbb{C}^r.$$

They have squared norm one, their Gram matrix is $I + T_{\text{pav}}$, and, with $d = 2/r$,

$$\sum_{v \in Q} \mathbb{E} u_v u_v^* = (D_0/r) \otimes I_r \leq dI.$$

Write the positive defect as a finite sum of deterministic outer products. Splitting its positive eigenvalues into finitely many summands at most one makes each added column have squared norm at most one. For all original and added columns together, their covariances A_j satisfy $\sum_j A_j = dI$ and $\text{Tr } A_j \leq 1$.

Theorem 4.1 of [MSS15] identifies the expected characteristic polynomial of a sum of independent, finitely supported rank-one outer products with the mixed characteristic polynomial of the covariances. Theorem 5.1 of [MSS15] gives largest root at most $(1 + \sqrt{\epsilon})^2$ for positive semidefinite covariances summing to I with individual traces at most ϵ . Apply these results to A_j/d , with $\epsilon = 1/d$, and rescale the spectral variable. The expected characteristic polynomial of the enlarged sum has largest root at most

$$d(1 + \sqrt{1/d})^2 = 1 + 2\sqrt{d} + d. \quad (3.19)$$

We justify carefully the removal of the added columns. For independent, finitely supported columns $u_1, \dots, u_m$ in a finite-dimensional space, let $A_j = \mathbb{E} u_j u_j^*$ and let U be the matrix of columns. Then

$$F(Y) := \mathbb{E} \det(\text{diag}(Y) - U^* U) = \left(\prod_{j=1}^m (Y_j - \partial_{s_j}) \right) \det \left(I + \sum_{j=1}^m s_j A_j \right) \Big|_{s=0}. \quad (3.20)$$

Indeed, the coefficient indexed by a subset J of columns on either side is $(-1)^{|J|}$ times the expected principal Gram determinant $\mathbb{E} \det(U_J^* U_J)$. This follows first for deterministic rank-one matrices by determinant expansion, and then for the mean outer products by independence and multilinearity in the distinct matrices indexed by J .

The determinant on the right of (3.20) is stable in the variables s_j . If it had a null vector with all $\text{Im } s_j > 0$, taking the imaginary part of its quadratic form would force the vector into $\ker A_j$ for every j ; the identity term then forces the vector to be zero. If a polynomial p is stable, fixing all variables except s_j at upper-half-plane arguments gives

$$\frac{\partial_{s_j} p}{p} = \sum_k \frac{1}{s_j - \lambda_k}, \quad \text{Im } \lambda_k \leq 0.$$

The sum is zero when the degree is zero, and otherwise has negative imaginary part. Thus a new upper-half-plane variable Y_j cannot equal this logarithmic derivative, and $(Y_j - \partial_{s_j})p$ is stable. Repeating this observation, and then specializing the s_j to zero by the zero-free limit theorem, proves stability of F . The limit is not the zero polynomial, since its leading monomial $\prod_j Y_j$ has coefficient one. The same proof works for every principal set of columns.

Differentiating F in a diagonal variable deletes the corresponding column from the expected Gram polynomial. The monic affine argument of Lemma 3.2 therefore applies to the quotient of its univariate specialization by the deleted polynomial. It has imaginary part at least $\text{Im } z$, and its polynomial part is $z - \mathbb{E} \|u_j\|^2$. The residue argument in that Lemma realizes the full polynomial as the characteristic polynomial of a Hermitian arrow matrix whose principal compression has the deleted polynomial as characteristic polynomial. The largest eigenvalue of a principal compression cannot exceed the largest eigenvalue of the whole matrix. Thus deleting columns cannot increase the largest root.

If the ambient column space has dimension q , the exact identity

$$z^q \mathbb{E} \det(zI_m - U^*U) = z^m \mathbb{E} \det(zI_q - UU^*)$$

shows that the enlarged Gram polynomial and the enlarged sum polynomial differ only by zero roots. Since the bound (3.19) is positive, it also bounds the largest root of the enlarged Gram polynomial. Delete every added column using the preceding paragraph. The original Gram polynomial is $\mathbb{E} \det(zI - (I + T_{\text{pav}})) = f_Q(z - 1)$. Shifting by minus one proves (3.18). A common penalty b replaces $f_Q(z)$ by $f_Q(z + b)$, giving the final assertion. $\square$

3.3 Residue tails and exceedance lengths

The rest of this section concerns arbitrary arrow quotients, not just those arising from a coloring. Let $s_j \in \mathbb{R}$, $\nu_j \geq 0$, and

$$h(z) = z + a - \sum_j \frac{\nu_j}{z - s_j}, \quad a > 0. \quad (3.21)$$

Let γ be the first-coordinate spectral probability measure of the corresponding arrow matrix, and put

$$m = \int x_+ d\gamma(x), \quad L(a, \nu) = \left| \left\{ \lambda > 0 : \sum_j \frac{\nu_j}{\lambda - s_j} > \lambda + a \right\} \right|. \quad (3.22)$$

Here and below $|\cdot|$ denotes Lebesgue measure when applied to a set of real numbers. By Lemma 3.2, this length is the positive-axis integral of the spectral shift for h .

Lemma 3.4 (Control of the residues). *Suppose $a = 1$ in (3.21), and set $H = \int |x| d\gamma(x) = 1 + 2m$. There is a universal C_0 such that*

$$\sum_{|s_j| \leq 60H} \nu_j \leq C_0 H^2, \quad (3.23)$$

$$\sum_{s_j \geq 20H} \frac{\nu_j}{s_j} \leq C_0 m, \quad (3.24)$$

$$\sum_{s_j \leq -20H} \frac{\nu_j}{|s_j|} \leq C_0 H. \quad (3.25)$$

For example, $C_0 = 6000$ suffices.

Proof. The first spectral moment is -1 , so $H = 1 + 2m \geq 1$. By Markov's inequality,

$$\gamma([-4H, 4H]) \geq 3/4.$$

Writing $G = 1/h$, we obtain

$$-\operatorname{Im} G(i10H) = \int \frac{10H}{x^2 + 100H^2} d\gamma(x) \geq \frac{15}{232H}.$$

Since $\operatorname{Im}(1/G) \leq 1/(-\operatorname{Im} G)$,

$$10H + \sum_j \frac{10H\nu_j}{s_j^2 + 100H^2} = \operatorname{Im} h(i10H) \leq \frac{232}{15} H < 16H.$$

For $|s_j| \leq 60H$, the denominator is at most $3700H^2$. It follows that the left side of (3.23) is at most $2220H^2$.

Let e be the first coordinate vector of the arrow matrix B , and let e_j be the other coordinate vectors, so that $(B - s_j)e_j$ is a scalar multiple of e of modulus $\sqrt{\nu_j}$. Fix $\lambda \geq 4H$ and $s_j \geq 2\lambda$. Spectral calculus on $(-\infty, \lambda]$, which is separated from s_j , and $\langle e_j, e \rangle = 0$ give

$$\begin{aligned} \left| \langle e_j, \mathbf{1}_{(\lambda, \infty)}(B)e \rangle \right| &= \left| \langle e_j, \mathbf{1}_{(-\infty, \lambda]}(B)e \rangle \right| \\ &= \sqrt{\nu_j} \int_{x \leq \lambda} \frac{d\gamma(x)}{s_j - x} \geq \frac{\sqrt{\nu_j}}{2s_j}. \end{aligned}$$

The last inequality uses $\gamma([-\lambda, \lambda]) \geq 3/4$ and $s_j + \lambda \leq 3s_j/2$. Bessel's inequality now yields

$$\sum_{s_j \geq 2\lambda} \frac{\nu_j}{s_j^2} \leq 4\gamma((\lambda, \infty)).$$

Integrating for $\lambda \geq 4H$ gives

$$\sum_{s_j \geq 8H} \frac{\nu_j}{s_j^2} \left(\frac{s_j}{2} - 4H \right) \leq 4 \int_{4H}^{\infty} \gamma((\lambda, \infty)) d\lambda \leq 4m.$$

For $s_j \geq 20H$ the summand on the left is at least $3\nu_j/(10s_j)$. Thus (3.24) holds with constant $40/3$. The same positive-tail calculation applied to $-B$ has first absolute moment H and positive first moment $\int (-x)_+ d\gamma(x) = 1 + m \leq H$. It gives (3.25) with the same constant. These calculations include zero residues and repeated poles. $\square$

The next lemma is Boole's identity for finite positive atomic measures [Boole1857]; see [AW15, Proposition 2.6 and Appendix A]. For this one-sided normalization, see [PSZ09, Section 1, equations (1.1), (1.3), and (1.5)]. Its elementary root-sum proof will also fix the sign and endpoint conventions used here.

Lemma 3.5 (Boole length identity). *For finitely many real poles s_j , nonnegative weights w_j , and $c > 0$,*

$$\left| \left\{ \lambda \in \mathbb{R} : \sum_j \frac{w_j}{\lambda - s_j} > c \right\} \right| = \left| \left\{ \lambda \in \mathbb{R} : \sum_j \frac{w_j}{\lambda - s_j} < -c \right\} \right| = \frac{\sum_j w_j}{c}. \quad (3.26)$$

If every pole carrying positive weight is strictly positive, the first exceedance set lies in $(0, \infty)$.

Proof. Merge coincident poles and omit zero weights. If there are no remaining poles the assertion is immediate. Otherwise list them in increasing order. The rational function is strictly decreasing between poles. Immediately to the right of each pole it tends to $+\infty$; before the next pole it tends to $-\infty$, and after the last pole it tends to zero. Thus the positive-level exceedance consists of one interval from each pole to the ensuing root of the level equation. Clearing denominators in $\sum_j w_j/(\lambda - s_j) = c$, comparison of the two highest coefficients shows that the sum of the level roots exceeds the sum of the poles by $(\sum_j w_j)/c$. This is precisely the total interval length. Reflection of the variable gives the negative-level identity. With positive poles the rational function is nonpositive on $(-\infty, 0]$, proving the last assertion. Values at the finitely many poles and endpoints do not affect these lengths. $\square$

3.4 The buffered choice

The buffer makes it possible to retain coefficient one on the existing cost ℓ . When m is large, we bound the whole child cost by a multiple of m ; when m is small, we reserve half of the existing length and control the remaining exceedance by the residue tails.

Lemma 3.6 (Buffered pin inequality). *There are universal constants $K \geq 1$ and $C > 0$ with the following property. In (3.21), let a finite family of choices have masses $\nu_{ij} \geq 0$ and probabilities $t_i \geq 0$ such that $\sum_i t_i = 1$ and $\sum_i t_i \nu_{ij} = \nu_j$. Keep the poles s_j unchanged, and define*

$$h'_i(z) = z + Ka - \sum_j \frac{\nu_{ij}}{z - s_j}, \quad \ell'_i = L(Ka, \nu_i), \quad \ell = L(a, \nu).$$

Then

$$\min_{i: t_i > 0} \ell'_i \leq \ell + Cm. \quad (3.27)$$

The constants have no dependence on the number or positions of the poles, the masses, the probabilities, or the positive penalty a .

Proof. Dividing the arrow matrix by a sends s_j to s_j/a , the masses to ν_j/a^2 , and both lengths and m to their values divided by a . It therefore suffices to prove the assertion for $a = 1$. Write $L = L(1, \nu)$ and $H = 1 + 2m$, and fix C_0 from Lemma 3.4. Only positive-probability choices are used below.

Suppose first that $m \geq 1$. The means of the two nonnegative statistics

$$A_i = \sum_{|s_j| \leq 20H} \nu_{ij}, \quad B_i = \sum_{|s_j| > 20H} \frac{\nu_{ij}}{|s_j|}$$

are at most $C_0 H^2$ and $C_0(m + H) \leq 2C_0 H$, respectively. Markov's inequality and a union bound give a choice satisfying

$$A_i \leq C_2 H^2, \quad B_i \leq C_2 H, \quad C_2 = 8C_0. \quad (3.28)$$

For $\lambda > DH$ and $D \geq 40$, all denominators from poles $|s_j| \leq 20H$ are positive and their total contribution is at most $C_2 H^2/(\lambda - 20H) \leq C_2 H$. The contribution from poles $s_j < -20H$ is at most $C_2 H$ as well. Put $C_3 = 2C_2$. For the remaining positive poles the identity

$$\sum_{s_j > 20H} \frac{\nu_{ij}}{\lambda - s_j} = (\lambda + H) \sum_{s_j > 20H} \frac{\nu_{ij}}{(s_j + H)(\lambda - s_j)} - \sum_{s_j > 20H} \frac{\nu_{ij}}{s_j + H} \quad (3.29)$$

holds. Choose $D \geq \max\{40, 2C_3 + 1\}$. For any $K \geq 1$, if the entire rational sum exceeds $\lambda + K$, then (3.29) forces

$$\sum_{s_j > 20H} \frac{\nu_{ij}}{(s_j + H)(\lambda - s_j)} > \frac{1}{2}.$$

Indeed the necessary lower bound for this sum is at least $(\lambda - C_3 H)/(\lambda + H) \geq 1/2$. By Lemma 3.5 and (3.28), its exceedance length is at most

$$2 \sum_{s_j > 20H} \frac{\nu_{ij}}{s_j + H} \leq 2C_2 H.$$

Including the initial interval $(0, DH)$, we obtain

$$L(K, \nu_i) \leq (D + 2C_2)H \leq 3(D + 2C_2)m. \quad (3.30)$$

This proves the required bound in this case, with a universal constant, for every $K \geq 1$.

It remains to treat $0 \leq m < 1$, so $1 \leq H \leq 3$. Set

$$P = \sum_{s_j > 0} \frac{\nu_j}{1 + s_j}, \quad M_-(d) = \sum_{s_j \leq 0} \frac{\nu_j}{d + |s_j|} \quad (d > 0),$$

using the right limit $M_-(0+)$ when necessary. Splitting the positive poles at 60 and using Lemma 3.4 gives

$$P \leq 9C_0 + C_0 m \leq C_4, \quad C_4 = 10C_0. \quad (3.31)$$

Here $60 \leq 60H$ controls the near part and $60 \geq 20H$ controls the tail. For $\lambda > 0$ away from the poles, the parent exceedance condition is equivalent to

$$R(\lambda) := \sum_{s_j > 0} \frac{\nu_j}{(1 + s_j)(\lambda - s_j)} > 1 + \frac{P - M_-(\lambda)}{1 + \lambda}. \quad (3.32)$$

The threshold on the right is at most $1 + P$. All poles of R with positive weight are positive. Lemma 3.5 therefore gives

$$L \geq \frac{P}{1 + P}, \quad P \leq C_5 L, \quad C_5 = 1 + C_4. \quad (3.33)$$

We next prove the uniform bound

$$M_-(L/2) \leq C_6, \quad (3.34)$$

with the right-limit interpretation when $L = 0$. If $L \geq 1$, split the negative poles at absolute value $60H$. The near part is at most $2C_0 H^2 \leq 18C_0$, while the far part is at most $C_0 H \leq 3C_0$. Thus

$21C_0$ suffices in this case. If $L = 0$, then (3.33) implies $P = 0$. If $M_-(0+) > 1$, including an infinite right limit, then $M_-(\lambda) > 1 + \lambda$ on a nonempty interval immediately to the right of zero. With no positive-pole contribution this would be an interval of parent exceedance, contradicting $L = 0$. Hence $M_-(0+) \leq 1$.

Finally suppose $0 < L < 1$. If $M_-(2L) > 7C_5 + 9$, then for $0 < \lambda < 2L$ we have $P \leq C_5$, $M_-(\lambda) \geq M_-(2L)$, and $1 + \lambda \leq 3$. Consequently the right side of (3.32) is strictly less than $-2(C_5 + 1)$. By the negative-level identity in Lemma 3.5,

$$|\{\lambda \in \mathbb{R} : R(\lambda) < -2(C_5 + 1)\}| = \frac{P}{2(C_5 + 1)} < \frac{L}{2}.$$

Outside this exceptional set, the parent exceedance holds throughout $(0, 2L)$, apart from finitely many poles. Its length is therefore greater than $3L/2$, contradicting the definition of L . It follows that $M_-(2L) \leq 7C_5 + 9$. Termwise comparison of denominators gives $M_-(L/2) \leq 4M_-(2L)$. Thus (3.34) holds in all cases with, for instance,

$$C_6 = \max\{21C_0, 1, 28C_5 + 36\}. \quad (3.35)$$

For each choice define P_i and $M_{-,i}$ by replacing ν with ν_i . Averaging these nonnegative quantities, followed by Markov's inequality and a union bound, gives a single choice i with

$$P_i \leq 4P, \quad M_{-,i}(L/2) \leq 4C_6, \quad \sum_{s_j > 60} \frac{\nu_{ij}}{s_j} \leq 4C_0m. \quad (3.36)$$

There is no exception when one of the means is zero: the corresponding nonnegative statistic then vanishes at every positive-probability choice. When $L = 0$, finiteness of $M_-(0+)$ forces $\nu_j = 0$ at every pole $s_j = 0$. The nonnegative averaging identities then force $\nu_{ij} = 0$ at those poles for every positive-probability choice. The right-limit child statistics are therefore finite sums over strictly negative poles, and their averages equal the parent statistic. Thus the same selection argument applies at $L = 0$ and at $m = 0$.

For this choice put $P_i^{(K)} = \sum_{s_j > 0} \nu_{ij}/(K + s_j)$. The child's exceedance condition is

$$\sum_{s_j > 0} \frac{\nu_{ij}}{(K + s_j)(\lambda - s_j)} > 1 + \frac{P_i^{(K)} - M_{-,i}(\lambda)}{K + \lambda}. \quad (3.37)$$

For $\lambda > L/2$ the negative statistic is at most $4C_6$ by monotonicity. Therefore $K \geq 8C_6$ makes the right side of (3.37) at least $1/2$. The length of the child exceedance beyond $L/2$ is at most $2P_i^{(K)}$ by Lemma 3.5. Split the positive poles at 60. For $0 < s_j \leq 60$,

$$\frac{1}{K + s_j} \leq \frac{61}{K} \frac{1}{1 + s_j},$$

whereas for $s_j > 60$ it is at most $1/s_j$. Thus

$$2P_i^{(K)} \leq \frac{122}{K} P_i + 2 \sum_{s_j > 60} \frac{\nu_{ij}}{s_j} \leq \frac{488C_5}{K} L + 8C_0m. \quad (3.38)$$

Choose once and for all

$$K \geq \max\{1, 8C_6, 976C_5\}.$$

The first term in (3.38) is at most $L/2$. Adding the possible length $L/2$ of the initial interval gives $L(K, \nu_i) \leq L + 8C_0m$. Together with (3.30), this proves (3.27) with $C \geq \max\{8C_0, 3(D + 2C_2)\}$. Scaling back to arbitrary $a > 0$ completes the proof. $\square$

We fix the universal constants K, C supplied by Lemma 3.6 for the remainder of the argument. By (3.17), at a vertex with positive penalty the lemma supplies a buffered pin increasing $\mathcal{E}(Q)$ by at most Cm_v . The coefficient one on ℓ_v is what cancels the old insertion cost in this difference. The measured argument will sum these local charges while making interactions between simultaneous pins negligible.

4 Measured matrices and coherent choices on Maharam slabs

Let $\mathcal{R}$ be a countable nonsingular Borel equivalence relation on a standard probability space (V, μ) . All measurable objects are understood modulo null sets, with Borel representatives. Our convention for the multiplicative Radon–Nikodym cocycle is

$$d\mu(w) = \omega(v, w) d\mu(v) \quad (4.1)$$

along partial isomorphisms $v \mapsto w$ with graph in $\mathcal{R}$. An invariant null set can be removed so that the cocycle identities hold on the resulting relation.

Consider a measurable system T of Hermitian contractions with zero diagonal on the spaces $\ell^2([v]_{\mathcal{R}})$. The system is *principal*: the indices are the points of an equivalence class, with one index for each point. On changing the root within a class, the matrices agree up to conjugation by diagonal unitaries. Suppose their entries are supported on an undirected Borel graph $\mathcal{G}$ in $\mathcal{R}$, of degree at most a fixed finite number D , and suppose that for a fixed $J \geq 1$,

$$J^{-1} \leq \omega(v, w) \leq J \quad \text{on every edge of } \mathcal{G}. \quad (4.2)$$

A measurable coloring of V defines T_{pav} by retaining entries whose two indices have the same color and setting the other entries to zero. For a function of an orbit matrix, its diagonal entry at the root v is denoted by a subscript vv .

Theorem 4.1 (One-sided measured paving). *There is a universal constant $K \geq 1$ with the following property. Suppose T , $\mathcal{R}$, $\mathcal{G}$, and ω satisfy the preceding assumptions. Let $b > 0$, and choose a positive integer r such that*

$$2\sqrt{2/r} + 2/r < b. \quad (4.3)$$

For every $\sigma > 0$ and $\rho > 0$, there is a measurable coloring of V with $r + 1$ colors for which

$$\int_V ((T_{\text{pav}} - (Kb + \sigma)\mathbf{1})_+^2)_{vv} d\mu(v) < \rho. \quad (4.4)$$

We use the constants K and C of Lemma 3.6. This section constructs choices that are coherent along the height coordinate of the Maharam extension: the finite buffered inequality selects a color at a lifted vertex, but a paving of the base requires the same choice at every height above that vertex. The proof of Theorem 4.1 is completed by the sparse pinning and descent argument in the next section. All constants called uniform in this section may depend on D, J, r, b and the fixed matrix system, but not on a partial assignment or on the endpoints of a slab. Constants for a single fixed slab will be identified separately.

4.1 Invariant measure and weighted polynomial systems

For the Maharam extension, see [Maharam64] and the formula in [RS06, Section 2, before Theorem 2.1]; the multiplicative coordinate there is $y = e^{-s}$ in our convention. We verify invariance in

the present cocycle notation before developing the slab construction. The Maharam extension has vertices $u = (v, s) \in V \times \mathbb{R}$ and relates (v, s) to

$$(w, s + \log \omega(v, w)), \quad (v, w) \in \mathcal{R}.$$

Projection to V is injective on each lifted orbit: the cocycle identities fix the height over w uniquely. The measure

$$d\tilde{\mu}(v, s) = d\mu(v)e^{-s} ds$$

is invariant along this relation. Indeed, writing $s' = s + \log \omega(v, w)$, Equation (4.1) gives

$$d\mu(w)e^{-s'} ds' = \omega(v, w) d\mu(v) \frac{e^{-s}}{\omega(v, w)} ds = d\mu(v)e^{-s} ds.$$

Consequently the mass-transport identity is

$$\int \sum_y F(u, y) d\tilde{\mu}(u) = \int \sum_u F(u, y) d\tilde{\mu}(y) \quad (4.5)$$

for nonnegative measurable transports along the lifted relation, and for integrable complex transports. The identity remains true after restricting both endpoints to a slab or to any further measurable subset. To justify it, decompose the countable Borel relation into countably many graphs of partial Borel isomorphisms, using the countable-section decomposition theorem and its application to the inverse relation. On each lifted graph it is precisely the preceding change of variables; summing the graphwise identities proves the claim. The scalar polynomial data used below are invariant under the allowed diagonal-unitary gauges, so they define transports without any choice of gauge.

Fix a bounded interval I and restrict the lifted graph and matrices to $V \times I$. A partial assignment on the base V specifies a color at some vertices. At unassigned vertices let $t_{vi} = 1/r$ for every i and let $c_v = b$; at assigned vertices let $(t_{vi})_i$ be the corresponding point mass and let $c_v = Kb$. Lift these data unchanged in s , and set

$$d_u = e^s \quad (u = (v, s)), \quad W = d^{1/2} T d^{1/2}, \quad a_u = d_u c_v. \quad (4.6)$$

The weight $d_u = e^s$ cancels the density e^{-s} of the invariant measure: an absolute spectral cost integrated against $d\tilde{\mu}$ is its normalized cost, divided by d_u , integrated against $d\mu ds$. This will make long slabs comparable by ordinary averages over height. We use the finite mixed polynomials and local data of Lemma 3.2 with these matrices and penalties. A prospective buffered pin at an unassigned lifted vertex fixes its color and replaces its penalty bd_u by Kbd_u . During comparisons we also allow arbitrary vertex deletions and single lifted-vertex changes; these intermediate configurations need not arise from a base assignment. Distances are always taken in the original restricted lift of $\mathcal{G}$, including edges whose entries have subsequently been set to zero.

All finite principal systems in a fixed slab, and all their prospective pins, have their roots and poles in a common interval $[-B, B]$. For example, an enlarged value of $\sup_u a_u + r \|W\|$ is sufficient: apply Lemma 3.1 to the two half-planes beyond the numerical ranges of the determinant matrices. The same conclusion holds if some cross edges are cut or interpolated as below, with an enlarged fixed-slab bound.

There are also estimates which do not depend on the slab. The residue formula in Lemma 3.2, together with $d_y/d_u \leq J$ on edges, gives

$$a_u \leq Kbd_u, \quad \sum_j \nu_{u,j} \leq \sum_y |W_{uy}|^2 \leq Jd_u^2 \sum_y |T_{uy}|^2 \leq Jd_u^2.$$

They hold for each pin alternative as well. They continue to hold under the cuts and real edge interpolations used later, since these only decrease the absolute values of the entries. The second-moment formula for γ_u and the arrow representation therefore give a constant C_{loc} such that

$$m_u \leq C_{\text{loc}} d_u, \quad 0 \leq \ell_u \leq C_{\text{loc}} d_u, \quad 0 \leq \ell'_{u,i} \leq C_{\text{loc}} d_u. \quad (4.7)$$

For the last two estimates, adding the off-diagonal arrow to $(-a_u) \oplus S$ changes the trace of the positive part by at most the trace norm of that arrow, namely $2(\sum_j \nu_{u,j})^{1/2}$. The positive-part trace variational formula proves this estimate; interlacing gives its nonnegative sign. Here and below h'_i denotes the buffered pin, rather than a derivative.

4.2 Locality and infinite bounded slabs

Lemma 4.2 (Local limits on a bounded slab). *On each bounded slab, the finite local functions $h_u, G_u, h_{u,i}, h'_{u,i}, R_{u,i}$, the probability and residue measures, and the data $m_u, \ell_u, \ell'_{u,i}$ have unique limits as finite graph balls increase to a component. They are measurable and independent of the exhaustion, and the local Cauchy-transform and Poisson-transform identities pass to the limits. Their supports have a common fixed-slab bound, and Equation (4.7) continues to hold.*

The Laurent coefficients through order k of h_u/z , zG_u , $h_{u,i}/z$, $h'_{u,i}/z$, and $R_{u,i}$, and the moments of γ_u through order k , depend only on the data within graph distance k of u . The same assertion holds for coefficients of logarithmic derivatives under data perturbations, with the local perturbation directions included in the data. It remains valid under arbitrary deletion configurations.

At an unassigned vertex the buffered pin inequality of Lemma 3.6 holds for these infinite local data. If every base vertex is unassigned, then $m_u = 0$ on every bounded slab.

Proof. We first establish locality directly at the level of formal Laurent series, so that the assertion does not depend on any limiting argument. For a finite principal set Q and $v \in Q$, put $H_{v,Q} = h_{v,Q}/z$. Expanding the determinant expectation according to the singleton or nontrivial permutation cycle containing v gives

$$H_{v,Q} = 1 + \frac{a_v}{z} - \sum_{\substack{C \ni v \\ |C| \geq 2}} w_C z^{-|C|} \prod_{x \in C \setminus \{v\}} H_{x,Q(x)}^{-1}. \quad (4.8)$$

Here w_C is the cycle weight in Equation (3.4). Delete v first and then the other vertices of C in a fixed order; $Q(x)$ denotes the principal set remaining just before x is deleted. The inverse quotients in the product telescope to $f_{Q \setminus C}/f_{Q \setminus \{v\}}$ after the displayed powers of z are accounted for. Each oriented permutation cycle is counted once, with the convention of the finite expansion.

All series H have constant term 1. A coefficient of order at most k in a term coming from a cycle of length l uses only coefficients through order $k-l$ in its inverse factors. Induction on k now proves the radius- k locality assertion: the cycle has length at most k , and the further radii of the needed factors stay within distance k of v . Formal inversion, multiplication, and differentiation preserve this property. This proves locality for all the listed series and for their perturbation derivatives, including complex perturbations; only the algebraic finite polynomial identities were used. The expansion of zG_v is the moment expansion of γ_v , proving the moment assertion.

For real data, the measures γ_v are probability measures with support in the common compact interval $[-B, B]$. The residue measures have support there and bounded mass for each fixed vertex, uniformly in the finite principal subset. Locality stabilizes all their moments: for the residue measure use the Laurent expansion of $h_v(z) = z + a_v - \int (z-x)^{-1} d\nu_v(x)$. Compact-support moment determinacy proves unique weak convergence. It also gives locally uniform convergence of h_v, G_v

and their pin alternatives on $\mathbb{C} \setminus [-B, B]$, in particular on the upper half-plane. Their limits satisfy $\text{Im } h_v(z) \geq \text{Im } z$.

For the finite spectral-shift functions ξ_h , we have $0 \leq \xi_h \leq 1$. Every weak-* subsequential limit therefore has these bounds. The identity

$$\pi^{-1} \arg h(x + i\eta) = (P_\eta * \xi_h)(x)$$

passes to the limit, since the Poisson kernel belongs to $L^1(\mathbb{R})$ and the analytic functions converge. Poisson approximation uniquely determines the limit function from these identities. Thus the entire sequence converges weak-*; all functions vanish above B , so their integrals over the positive axis converge. The integrals defining m also converge by the compact support of γ . Apply this reasoning to every pin alternative. The inequalities in Equation (4.7) and Lemma 3.6 pass to the limit, the latter because there are only finitely many choices of color.

Finite graph balls and the corresponding finite polynomial computations are measurable in the countable Borel relation. Their limits are therefore measurable. The same construction applies to single-vertex response functions indexed by pairs in an orbit. Components disconnected in the distance graph do not affect these local functions, so they can be treated separately. Throughout this construction the heights and weights are absolute on the lifted orbit; they are not reset when the root changes.

Finally suppose all vertices are unassigned. On a finite set Q , the weighted polynomial at common argument z is $\prod_{v \in Q} d_v$ times the unweighted mixed polynomial for T , with zero penalties, evaluated at the tuple $(b + z/d_v)_{v \in Q}$. Its multiaffine Taylor coefficients at the constant tuple b are the corresponding smaller principal polynomials evaluated at b . All are strictly positive by Equation (4.3) and Lemma 3.3, including the empty polynomial 1. Hence the weighted polynomial is strictly positive for every real $z \geq 0$. All its roots are real, so its local spectral measures have no positive support. Taking the local limits gives $m_u = 0$. $\square$

4.3 Uniform comparison under far deletions

At an unassigned vertex of a slab define the normalized tuple

$$\mathcal{D}^I(v, s) = e^{-s} (m_{(v,s)}, \ell_{(v,s)}, (\ell'_{(v,s),i})_{1 \leq i \leq r}). \quad (4.9)$$

We use the sum of absolute values as its norm. The superscript records the slab, and will be omitted when the ambient configuration is clear.

Lemma 4.3 (Far-deletion comparison). *Consider any bounded slab containing $V \times [0, 1]$, any base partial assignment, and any further measurable restriction that retains the central vertices being tested. Deleting a measurable set of vertices at heights $s > R$, or at heights $s < -R$, changes the tuple in Equation (4.9) by an amount tending to zero in $L^1(d\mu ds)$ on the central unassigned band as $R \rightarrow \infty$. The convergence is uniform in the slab, the partial assignment, and the deletion set. The same conclusion holds for the union of a far high set and a far low set.*

Proof. We prove the analytic comparisons first and then recover the integral data. The transform G determines the local probability measure; $h = 1/G$ and $R_i = h_i/h$ then determine the parent and pinned spectral shifts. For high deletions we test their integrated differences against bounded complex directions and express a single deletion response as a logarithmic derivative. Stability bounds that derivative independently of height. We transport its local Laurent coefficients and use analytic uniqueness to recover the whole-deletion response. Low deletions instead cut a matrix of small norm. All pre- and post-deletion systems use the same absolute heights, penalties, and retained probabilities.

High deletions and test directions. Write D for a set above height R , with $R > 1$. Then $\tilde{\mu}(D) \leq e^{-R}$. For upper-half-plane arguments z we claim

$$\int_{\text{central}} \left| G_v^{\text{pre}}(z) - G_v^{\text{post}}(z) \right| d\tilde{\mu}(v) \leq C(z)\tilde{\mu}(D), \quad (4.10)$$

$$\int_{\substack{\text{central} \\ \text{unassigned}}} \left| R_{v,i}^{\text{pre}}(z) - R_{v,i}^{\text{post}}(z) \right| d\tilde{\mu}(v) \leq C(z)\tilde{\mu}(D). \quad (4.11)$$

Here v denotes a lifted vertex, and $R_{v,i} = h_{v,i}/h_v$ uses the pin with unchanged penalty. The constants are bounded uniformly for z in upper-half-plane compact sets, independently of the slab and assignment.

It suffices to bound the integral of each complex difference against an arbitrary measurable direction $\lambda(v)$ with $|\lambda(v)| \leq 1$, supported on the tested set. For the first comparison perturb Y_v by $\zeta\lambda(v)$. For the second, fix a color i and add $\zeta\lambda(v)$ to its probability at every tested unassigned vertex, leaving the other probabilities unchanged. For this second perturbation, first use unnormalized weights in the polynomial sum. Equivalently, normalize the weights at each vertex and multiply the resulting polynomial on Q by $\prod_{v \in Q} (1 + \zeta\lambda(v))$. These additional factors cancel in the quotient deleting one high vertex u , since $\lambda(u) = 0$.

In a finite principal system, differentiation of the numerator and denominator yields

$$U_u(z) := \partial_\zeta \log h_u(z; \zeta)|_{\zeta=0} = \sum_{v \text{ tested}} \lambda(v) (\mathcal{B}_v^{\text{pre}}(z) - \mathcal{B}_v^{\text{post}(u)}(z)), \quad (4.12)$$

where $\mathcal{B}_v = G_v$ or $R_{v,i}$, respectively, and $\text{post}(u)$ means deletion of this one vertex. For the probability perturbation, the unnormalized derivative at v is exactly the polynomial with v pinned to color i ; its logarithmic derivative is $R_{v,i}$. The cancellation just noted allows us to estimate $h_u(z; \zeta)$ using the normalized determinant formula in Lemma 3.1.

A uniform bound for the logarithmic derivative. Fix $\eta > 0$ and upper arguments with $\text{Im } z \geq \eta$. There is a disk $|\zeta| < s_*$, independent of the slab, directions, finite principal set, and assignment, on which $\text{Im } h_u(z; \zeta) \geq \eta/2$. For the diagonal perturbation, the relevant determinant matrices change by norm at most $|\zeta|$. For the probability perturbation, all tested vertices initially have probabilities $1/r$. The normalized probabilities have analytic square roots near these positive values on a common disk $|\zeta| < c/r$, and their changes are bounded by a constant times $|\zeta|$ on a smaller disk. Square roots at all other vertices are unchanged, including the zero probabilities at assigned vertices. Every changed matrix entry is incident to the central band, and its other endpoint has height in $[-\log J, 1 + \log J]$ by Equation (4.2). Bounded row and column sums, using the degree bound, therefore bound the entire matrix perturbation by $C|\zeta|$, uniformly in the enclosing slab.

The initial determinant matrices are Hermitian. Taking s_* small enough, their perturbed numerical ranges have imaginary parts less than $\eta/2$ in absolute value. Lemma 3.1 gives nonvanishing when all diagonal variables have imaginary part greater than $\eta/2$, for every principal subset. In the diagonal-perturbation case, regard the shifts as perturbations of the penalties. The final quotient at the unperturbed vertex u is monic affine in Y_u ; its zero therefore has imaginary part at most $\eta/2$. Evaluating at the common argument z proves the asserted lower bound on the imaginary part of $h_u(z; \zeta)$.

A holomorphic logarithm on the ζ disk now has imaginary part between 0 and π . The harmonic gradient estimate and the Cauchy–Riemann equations give

$$|U_u(z)| \leq C(\eta). \quad (4.13)$$

The bound does not involve the size of h_u or the height of u . Thus high vertices with arbitrarily large real weights cause no loss of uniformity.

For each fixed bounded enclosing slab, we also need uniform control near infinity. On a common small ζ disk all perturbed determinant matrices then have a uniform norm bound. Choose B' larger than this bound and larger than every real support bound, with positive slack. For $|z| > B'$, use the half-plane in the direction $z/|z|$ strictly beyond the matrix norm bound. The affine quotient belongs to a rotated half-plane through the origin, so the same logarithmic argument gives a uniform bound for U_u there. This bound and B' may depend on the fixed slab; they are uniform in its principal subsets, directions, and assignment states. In particular the functions U_u are analytic and locally bounded on

$$\Omega = \mathbb{C}^+ \cup \{z : |z| > B'\}, \quad (4.14)$$

and are bounded near the removable point at infinity.

As finite balls around u increase, normal families give subsequential locally uniform limits on Ω , also near infinity. The locality in Lemma 4.2, applied to perturbation derivatives, stabilizes every Laurent coefficient at infinity. All subsequential limits therefore have the same Laurent series and coincide by the identity theorem on the connected domain Ω . This defines a unique measurable function U_u for infinite components, retaining Equation (4.13).

The endpoint functions $\mathcal{B}_v$ are also analytic on Ω , with removable values at infinity. At central vertices they are uniformly bounded on upper-half-plane compact sets. Indeed the residue representation gives

$$|h_{v,i}(z)| \leq |z| + a_v + \frac{\sum_j \nu_{v,i,j}}{\operatorname{Im} z}, \quad \operatorname{Im} h_v(z) \geq \operatorname{Im} z,$$

and the central residue masses and penalties are uniformly bounded. The same statements hold after deletions.

Recovering the whole-deletion response. Let $H_*(z)$ be the integral against $\lambda(v)$ of the difference between the endpoint functions before and after deleting all of D . The preceding bounds show that H_* is analytic on Ω , including at infinity. We do not sum individual analytic responses over an entire orbit. Instead we first match their local Laurent coefficients.

For each integer $k \geq 1$, partition D measurably into finitely many batches such that distinct vertices of a batch have distance greater than $2k$ in the enclosing graph, and delete the batches in order. The required finite Borel coloring exists for every bounded-degree Borel graph and each finite-distance graph power [KST99]. For completeness, choose a countable family of Borel binary labels separating points; assign each vertex a finite prefix long enough to distinguish it from its finitely many neighbors. These prefixes give a countable proper Borel coloring. Process the countably many color classes in order, recoloring each class with a palette of size one more than the degree bound and avoiding the previously assigned neighbor colors. Each class is independent, so this is a proper finite Borel coloring.

For each batch use the single-deletion logarithmic derivative in the configuration just before that batch, and put

$$S_k(z) = \sum_{\text{batches}} \int_{\text{batch}} U_u^{\text{pre}}(z) d\tilde{\mu}(u).$$

Because the batches partition D , Equation (4.13) gives the locally uniform bound $|S_k(z)| \leq C(z)\tilde{\mu}(D)$ on $\mathbb{C}^+$, independent of k and of the number of batches. There is a common fixed-slab bound near infinity as well.

The Laurent coefficients of S_k and H_* agree through order k . To see this, first apply Equation (4.12) to one deletion. At these orders only vertices within distance k of the deletion can contribute, by Lemma 4.2; hence this coefficient identity on an infinite component is verified in a sufficiently large finite neighborhood. Within one batch the coefficient change at a tested vertex is the sum of the single changes in the pre-batch configuration, because at most one batch vertex belongs to its radius- k neighborhood. Sum this coefficient identity over the tested vertices and use Equation (4.5) to move the sum to the deletion vertices. The transports are integrable: for the fixed slab and fixed order their coefficients are bounded, and their supports have a uniform finite ball-size bound. The analytic bounds near infinity justify interchanging coefficients and integrals. Finally sum over the batches; the successive endpoint changes telescope to H_* .

Apply normal families to S_k on Ω with its removable point at infinity. Every subsequential limit has every Laurent coefficient of H_* , since the first k coefficients agree for each k . The identity theorem shows that this limit is H_* throughout Ω . It follows that $|H_*(z)| \leq C(z)\tilde{\mu}(D)$ on $\mathbb{C}^+$, with the slab-independent constants already established there. The fixed-slab compactness near infinity has thus transferred the uniform upper-half-plane bound; it has not supplied a spectral-support bound uniform over slabs. For each fixed target z , choosing the measurable complex sign λ of the endpoint difference proves Equations (4.10) and (4.11).

At central vertices h_v and $h_{v,i}$ are bounded uniformly on upper-half-plane compact sets. The identity $h^{\text{pre}} - h^{\text{post}} = h^{\text{pre}} h^{\text{post}} (G^{\text{post}} - G^{\text{pre}})$ therefore gives the same L^1 comparison for h . Then $h_i = h R_i$ gives the comparison for every h_i on the central unassigned set. Adding the unchanged shift $(K-1)a_v$ gives it for h'_i as well.

Low deletions. Suppose now that D lies below height $-R$. Cut every edge between D and its complement, writing V_0 for the removed matrix. Each such edge has both heights below $-R + \log J$. Its entry is therefore $O(e^{-R})$; the degree bound yields $\|V_0\| = O(e^{-R})$ uniformly in the slab. Interpolate by $W(t) = W_{\text{cut}} + tV_0$, $0 \leq t \leq 1$. For finite systems at a central vertex, $h_v(z; t)$ lies in the upper half-plane and is bounded uniformly for upper z in compact sets, by the central second-moment and residue estimates. Around each real t , allow complex perturbations in a disk of radius $c\eta/(r\|V_0\|)$ when $\|V_0\| > 0$, where $\text{Im } z \geq \eta$. The corresponding determinant matrix perturbation has norm at most $c\eta$. The complex interpolation still has zero diagonal, so the algebraic identity and conditional half-plane assertion of Lemma 3.1 apply even though the matrix need not be Hermitian away from real t . The stability and logarithmic gradient argument just given implies

$$|\partial_t \log h_v(z; t)| \leq C(\eta, r) \|V_0\|.$$

The zero-norm case is immediate. The argument applies also to each prospective buffered pin, considered as a separate real-data system under the same edge interpolation. Integrate in real t , using the uniform absolute bound on h and $\text{Im } h \geq \eta$ for reciprocation. We obtain $O(e^{-R})$ endpoint comparisons for h, G, h'_i at every tested central vertex. In the cut system the low component factors out of the polynomial quotients, so the endpoint at $t = 0$ has exactly the same central data as the deleted system. The finite-ball limits pass these comparisons to infinite bounded-slab components.

Recovering the integral data. At central vertices the probability measures γ_v have uniformly bounded second moments. Therefore $m_v = \int x_+ d\gamma_v$ is uniformly approximated by integration against continuous compactly supported cutoffs of x_+ . For such a test function, Poisson convolution approximates it uniformly, so its integral against γ_v is in turn uniformly approximated by its integral against $-\pi^{-1} \text{Im } G_v(x + i\eta) dx$ at a sufficiently small fixed $\eta > 0$.

For the ℓ data we also have a uniform positive-tail estimate:

$$\int_M^\infty \xi_h(x) dx \leq \frac{\sum_j \nu_j}{M}, \quad M > 0. \quad (4.15)$$

For finite systems, $\xi_h(x) = 1$ with $x > M$ implies $\sum_j \nu_j / (x - s_j) > x + a_v > M$. The Boole length identity used in Lemma 3.6 proves Equation (4.15); weak-* convergence on a fixed bounded slab gives the infinite version. The bound is uniform at central vertices and for their pins. After choosing M large, approximate the remaining integral over $(0, M)$ by

$$\frac{1}{\pi} \int_0^M \arg h(x + i\eta) dx.$$

By the Poisson identity, the error is at most $\|P_\eta * 1_{(0,M)} - 1_{(0,M)}\|_{L^1(\mathbb{R})}$, which tends to zero as $\eta \downarrow 0$. At this fixed height, arguments are Lipschitz-controlled by differences of h values in $\text{Im } h \geq \eta$.

Choose the tail and smoothing errors first; then use the preceding analytic comparisons on the resulting compact upper-half-plane sets and integrate in x . This proves convergence for m, ℓ and all the ℓ'_i . On the central band the measures $d\tilde{\mu}$ and $d\mu ds$, and the normalization factors e^{-s} , are boundedly comparable, proving the claimed tuple comparison. A union of far high and far low deletions is handled in two successive steps. Non-strict far cutoffs have the same conclusion, by inserting a unit of slack in R . $\square$

4.4 A color choice independent of height

Figure 1 summarizes how the far-deletion estimates allow a single base color to be used throughout a long slab.

Lemma 4.4 (Coherent choices). *For every $\tau_0 > 0$ there are arbitrarily large integers $L \geq 1$ with the following property, uniformly over all base partial assignments. On the slab $V \times [0, L]$ one can choose a measurable prospective color $i(v)$ at each unassigned base vertex, using the same color at every height, so that*

$$\frac{1}{L} \int_{\text{unassigned in } V \times [0, L]} (\ell'_{(v,s), i(v)} - \ell_{(v,s)}) d\tilde{\mu}(v, s) \leq C\Phi + \tau_0, \quad \Phi = \frac{1}{L} \int_{V \times [0, L]} m_{(v,s)} d\tilde{\mu}(v, s). \quad (4.16)$$

Here C is the universal constant in Lemma 3.6, and every $\ell'_{u,i}$ on the left is the insertion cost after changing only the lifted vertex u . The same L can be used at every stage of a sequence of partial assignments.

Proof. The normalized tuple obeys the exact translation rule

$$\mathcal{D}^I(v, s + \alpha) = \mathcal{D}^{I-\alpha}(v, s). \quad (4.17)$$

Indeed translating the heights by α multiplies all weights d , the matrix W , and the penalties in the orbit by e^α . The spectral quantities m, ℓ, ℓ'_i scale by this same factor, which is canceled by the normalization at the root. The lifted probabilities are unchanged because the partial assignment is on the base. This proves Equation (4.17) first in finite balls and then in the local limits.

Take a large integer R , use the reference interval $[-R, R]$, and write its tuple as $\mathcal{D}^0$ in this proof. For $\alpha \in [-1, 1]$, compare this interval and its translate $[-R, R] - \alpha$ to their intersection. The differences involve only far high and low deletions. Equations (4.17) and Lemma 4.3 imply

$$\int_{v \text{ unassigned}} \int_{\{s \in [0, 1]: s + \alpha \in [0, 1]\}} |\mathcal{D}^0(v, s + \alpha) - \mathcal{D}^0(v, s)| ds d\mu(v) \leq e(R), \quad (4.18)$$

where $e(R) \rightarrow 0$ uniformly in the assignment. We enlarge $e(R)$ if necessary so that it is positive. Integrating in α turns the left side into the corresponding integral over all pairs $(s, t) \in [0, 1]^2$. Fubini therefore gives a single $s_0 \in [0, 1]$ such that

$$\int_{v \text{ unassigned}} \int_0^1 |\mathcal{D}^0(v, s) - \mathcal{D}^0(v, s_0)| ds d\mu(v) \leq 3e(R). \quad (4.19)$$

The slice can also be chosen so that the local data and the buffered pin inequality hold almost everywhere on it.

At this slice choose the least minimizing color for $\ell'_{(v, s_0), i}$. This is a measurable function $i(v)$ because there are finitely many colors. Lemma 3.6 gives

$$e^{-s_0}(\ell'_{(v, s_0), i(v)} - \ell_{(v, s_0)} - Cm_{(v, s_0)}) \leq 0.$$

The tuple norm contains all color alternatives. Consequently Equation (4.19) implies that the integral of this same normalized difference over v and $s \in [0, 1]$, now using the data at s , is at most a fixed constant times $e(R)$.

Consider a unit band $[k, k+1]$ in $[0, L]$, where k is an integer with $k \geq R$ and $L - k \geq R$. Translate it to $[0, 1]$; its ambient interval becomes $[-k, L - k]$. Trimming this interval to $[-R, R]$ removes only far high and low vertices. Lemma 4.3 and Equation (4.17) show that its tuple differs from $\mathcal{D}^0(v, s)$ in integrated norm by at most $e(R)$, after enlarging the same uniformly vanishing error function. Thus the μds integral over this unit band of

$$e^{-s}(\ell'_{(v, s), i(v)} - \ell_{(v, s)} - Cm_{(v, s)})$$

on unassigned vertices is at most a constant times $e(R)$. All remaining boundary bands have total length at most $2R + 2$. Their normalized data are uniformly bounded by Equation (4.7).

Absolute costs against $d\tilde{\mu}$ are exactly normalized costs against $d\mu ds$. Summing the unit-band estimates and dividing by L bounds the total error by $C_1 e(R) + C_2(R + 1)/L$, with constants independent of the assignment. Choose R sufficiently large, then choose L sufficiently large and satisfying any prescribed lower bound. This makes the error at most τ_0 . Finally $m \geq 0$ permits replacing its integral over unassigned vertices by the integral over the whole slab, yielding Equation (4.16). All choices of the numerical parameters used only uniform estimates, proving the assertion about using the same L at different stages. Null modifications of base data can be absorbed into an invariant null set by nonsingularity. $\square$

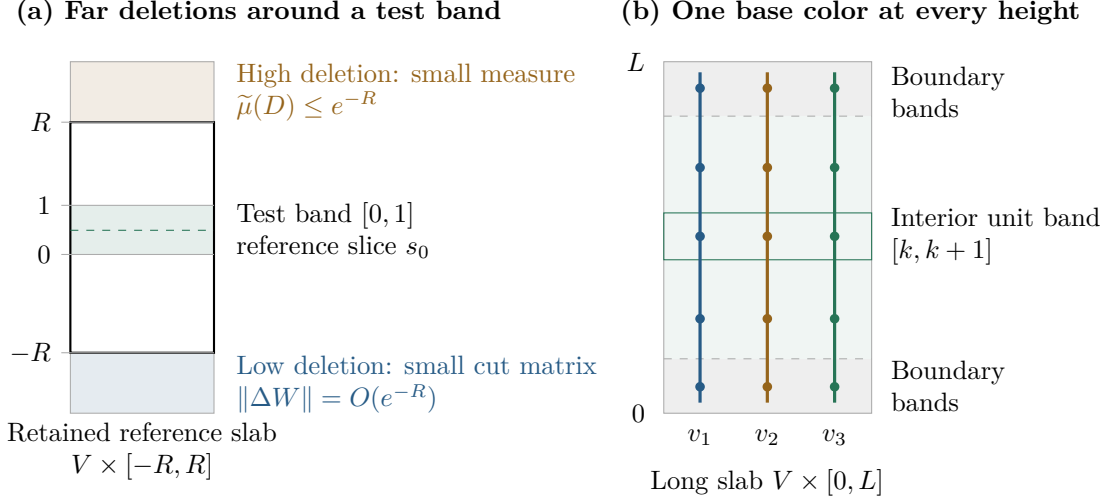


Figure 1: Height geometry behind Lemmas 4.3 and 4.4 (not to scale). High and low deletions are controlled by different estimates. A color selected on the reference slice is used at every height above the same base vertex. Translated interior unit bands compare with the reference slab in integrated normalized data; the boundary bands have total length at most $2R + 2$. The colored vertical lines are base fibers $\{v\} \times [0, L]$, not lifted relation orbits.

5 Sparse pinning and descent

5.1 Sparse pinning and removal of unassigned vertices

We retain the notation and hypotheses of Theorem 4.1. In particular, $b > 0$ and r satisfy its strict largest-root inequality, and K, C are the universal constants from Lemma 3.6. The lifted graph on a slab is always the restriction of the original enclosing graph, including edges at which a particular matrix entry happens to vanish. All distances below refer to this enclosing graph.

Proposition 5.1. *For every $\zeta, \eta > 0$ and $L_0 \geq 1$, there are an integer $L \geq L_0$ and a measurable partition*

$$V = \mathcal{U} \sqcup V_1 \sqcup \cdots \sqcup V_r$$

such that $\mu(\mathcal{U}) < \eta$ and the following holds. On the surviving lifted vertices $(V \setminus \mathcal{U}) \times [0, L]$, let T_{col}^L be the restriction of T , keeping only entries between vertices whose basepoints belong to the same V_i , and put

$$A_L = d^{1/2}(T_{\text{col}}^L - Kb\mathbf{1})d^{1/2}, \quad d_{(v,s)} = e^s. \quad (5.1)$$

Extend $m_u^{\text{surv}} = ((A_L)_+)_{uu}$ by zero on the deleted vertices. Then

$$\frac{1}{L} \int_{V \times [0, L]} m_u^{\text{surv}} d\tilde{\mu}(u) < \zeta. \quad (5.2)$$

Proof. We first construct a partial assignment with small mixed-polynomial cost. Polynomial moments depend on finite graph balls, so sufficiently sparse simultaneous pins have an additive first-order response except on an event of quadratic order in the update probability. Approximating the derivative of the positive-part function controls the error in that response independently of the number of roots. Parameters used to select the slab will be specified below. On a fixed slab $[0, L]$, write

$$\Phi = \frac{1}{L} \int_{V \times [0, L]} m_u d\tilde{\mu}(u).$$

Choose a common spectral support bound $[-B, B]$ for all the finite principal systems, pin alternatives, and deletion configurations on this slab, as in Lemma 4.2. For $\tau > 0$, choose a real polynomial P such that

$$P(-B) = 0, \quad \int_{-B}^B |P'(x) - \mathbf{1}_{(0,\infty)}(x)| dx \leq \tau. \quad (5.3)$$

For example, approximate the indicated step function first by a continuous function in $L^1([-B, B])$, then uniformly by a polynomial, and integrate that polynomial from $-B$. It follows that

$$\sup_{[-B, B]} |P(x) - x_+| \leq \tau.$$

Define

$$m_{P,u} = \int P(x) d\gamma_u(x), \quad \Phi_P = \frac{1}{L} \int_{V \times [0, L]} m_{P,u} d\tilde{\mu}(u).$$

Since γ_u is a probability measure and $\tilde{\mu}(V \times [0, L]) = 1 - e^{-L} \leq 1$, we have

$$|\Phi - \Phi_P| \leq \tau \quad (L \geq 1). \quad (5.4)$$

The response to one buffered pin. At an unassigned lifted vertex u , consider pinning to a specified color i , increasing its penalty from bd_u to Kbd_u at the same time. The total polynomial response on its component is

$$\begin{aligned} D_P(u) &= \sum_y (m_{P,y}^{\text{pin } u \text{ to } i} - m_{P,y}^{\text{pre}}) \\ &= \int_{-B}^B P'(x) (\xi_{h'_{u,i}}(x) - \xi_{h_u}(x)) dx. \end{aligned} \quad (5.5)$$

Here $h'_{u,i}$ denotes the buffered alternative; the prime does not denote differentiation. To prove the identity on a finite principal set, use the fact that the sum of its local spectral probability measures is the counting measure of the roots. The pre-pin and post-pin polynomials have equal degree and the same principal polynomial with u removed. Integrating the difference of their root counts above x against $P'(x)$ gives exactly (5.5).

Put $k = \max\{1, \deg P\}$. By the locality in Lemma 4.2, only vertices at distance at most k from u contribute to the sum on the first line of (5.5). Compute in finite balls large enough to contain these vertices and their relevant neighborhoods. The local polynomial moments stabilize, while the functions ξ converge weak-* on $[-B, B]$. This proves the identity on infinite components as well. Since both ξ -functions take values in $[0, 1]$ almost everywhere, (5.3) implies

$$|D_P(u) - (\ell'_{u,i} - \ell_u)| \leq \tau. \quad (5.6)$$

One sparse update. For a current partial base assignment, choose the prospective colors $i(v)$ supplied by Lemma 4.4, with tolerance $\tau_0 > 0$. The color $i(v)$ is the same at all heights over v . Properly color the distance- $2k$ power of the base graph with finitely many auxiliary Borel color classes. The finite Borel coloring fact used in the proof of Lemma 4.3 applies because this power still has bounded degree. Independently for these auxiliary classes, select a Bernoulli- α indicator, where $0 < \alpha < 1$. Every currently unassigned base vertex in a selected class is pinned to its prospective color, with its penalty buffered, at all heights simultaneously.

Within a radius- k lifted ball, different vertices project to different base vertices, and their pairwise base distance is at most $2k$. They consequently have different auxiliary classes. The indicators of

all candidate pins affecting the local moment $m_{P,y}$ are therefore independent. This assertion uses injectivity of projection within each lifted orbit; it does not require independent indicators at all vertices of the base.

If at most one candidate in this local ball is selected, the change in $m_{P,y}$ is exactly the sum of its selected single-pin responses. A ball has at most a fixed finite number N of vertices, so the probability of two or more selections is at most $\binom{N}{2}\alpha^2$. On this event the discrepancy between the actual change and the sum of single-pin responses is bounded in absolute value by a constant depending only on N and $\sup_{[-B,B]} |P|$. Integrate this estimate and exchange roots and pin locations using (4.5). The result is

$$\mathbb{E}[\Phi_P^{\text{new}} \mid \text{pre}] \leq \Phi_P + \frac{\alpha}{L} \int_{\text{unassigned}} D_P(u) d\tilde{\mu}(u) + C_P \alpha^2. \quad (5.7)$$

All the transports here are measurable, uniformly bounded on the fixed slab, and supported on bounded-radius pairs. Thus they are integrable and the signed version of mass transport is applicable. The constant C_P may depend on the slab, graph and polynomial, but not on the current assignment, the prospective choices, or α .

By Lemma 4.4, (5.6), and (5.7) gives

$$\mathbb{E}[\Phi_P^{\text{new}} \mid \text{pre}] \leq (1 + C\alpha)\Phi_P + \alpha(\tau_0 + (1 + C)\tau + C_P\alpha). \quad (5.8)$$

The same L from Lemma 4.4 works for all partial base assignments, so this step can be repeated.

Iteration and choice of parameters. Start with every vertex unassigned. The initial cost is $\Phi = 0$ by Lemma 4.2. Use fresh independent class indicators at each step and recalculate the prospective colors from the current assignment. Given $S > 0$, perform $n = \lceil S/\alpha \rceil$ steps. For these fixed parameters there are only finitely many possible histories. At every node one may choose separately the measurable prospective-color functions furnished by Lemma 4.4; no selection measurable in an infinite-dimensional space of assignments is needed.

We have $n\alpha \leq S + 1$ and $(1 + C\alpha)^n \leq e^{C(S+1)}$. Iterating (5.8), using the initial bound $\Phi_P \leq \tau$, and then using (5.4) again yields

$$\mathbb{E}\Phi_{\text{final}} \leq \tau + e^{C(S+1)}[\tau + (S + 1)(\tau_0 + (1 + C)\tau + C_P\alpha)]. \quad (5.9)$$

If $\mathcal{U}$ is the final unassigned base set, each still-unassigned vertex has conditional probability α of being selected at the next step, regardless of its prospective color. Therefore

$$\mathbb{E}\mu(\mathcal{U}) = (1 - \alpha)^n \leq e^{-S}. \quad (5.10)$$

Here is an order of choices which makes both expectations small. First take S so large that $e^{-S} < \eta/4$. Next take $\tau_0, \tau > 0$ so small that the terms in (5.9) not involving $C_P\alpha$ have sum less than $\zeta/8$. Choose an integer $L \geq L_0$ using Lemma 4.4 for this τ_0 . Now choose B and P as above, thereby fixing C_P . Finally take α small enough that the remaining term in (5.9) is less than $\zeta/8$. Thus

$$\mathbb{E}\Phi_{\text{final}} < \zeta/4, \quad \mathbb{E}\mu(\mathcal{U}) < \eta/4.$$

Both random variables are nonnegative. By Markov's inequality and a union bound, some history satisfies simultaneously

$$\Phi_{\text{final}} < \zeta, \quad \mu(\mathcal{U}) < \eta. \quad (5.11)$$

Fix such a history.

Removing the unassigned vertices. We show that deleting $\mathcal{U} \times [0, L]$ cannot increase the integrated positive cost. Keep the initial and final deletion states fixed. For an arbitrary $\vartheta > 0$, choose another polynomial Q with

$$Q(-B) = 0, \quad \int_{-B}^B |Q'(x) - \mathbf{1}_{(0,\infty)}(x)| dx \leq \vartheta.$$

At each intermediate deletion state, extend $m_{Q,y}$ by zero at vertices that have already been deleted.

For one deletion at u , the total pre-minus-post polynomial cost is

$$\sum_y (m_{Q,y}^{\text{pre}} - m_{Q,y}^{\text{post}}) = \int_{-B}^B Q'(x) \xi_{h_u}(x) dx \geq -\vartheta. \quad (5.12)$$

For finite sets the identity again follows by integrating the difference of root counts; the boundary term caused by the loss of one root is $Q(-B) = 0$. The inequality uses $0 \leq \xi_{h_u} \leq 1$. The finite-range locality argument used for (5.5) proves the identity in an infinite component.

Partition the set to be deleted into finitely many Borel batches whose distinct vertices have distance greater than $2 \max\{1, \deg Q\}$ within each batch in the enclosing lifted graph. For one batch, at most one deletion affects any local polynomial moment, so the batch change is exactly the sum of single-deletion responses in its pre-state. Mass transport, (5.12), and telescoping over the batches show that the integrated pre-minus-post polynomial cost, divided by L , is at least

$$-\vartheta \frac{\tilde{\mu}(\mathcal{U} \times [0, L])}{L}.$$

The error is charged once to each deleted vertex; it is not multiplied by the number of batches. Intermediate deleted sets need not be constant in the height variable, since Lemma 4.2 allows arbitrary deletion configurations in the enclosing slab.

At each endpoint the integrated error between the Q -cost and the positive cost is at most ϑ . Thus the pre-minus-post positive cost is at least -3ϑ . The endpoints are independent of Q and of the batch decomposition, so letting $\vartheta \downarrow 0$ proves the desired monotonicity.

All surviving vertices are now deterministically colored, with penalty Kbd_u . Their finite mixed polynomial is the characteristic polynomial of the corresponding restriction of (5.1), so its local measure is the diagonal spectral probability measure of that operator. On an infinite component the same assertion follows from stabilization of polynomial moments and the common compact spectral support. Hence the surviving positive cost is exactly m_u^{surv} . Monotonicity and (5.11) give (5.2). $\square$

5.2 Descent from a slab to the base relation

The next estimate converts the integrated weighted cost into the desired spectral error on the base. A finite ball with a large eigenvalue forces a definite positive cost in its lift; mass transport then bounds the measure of such bad balls. For an integer $k \geq 1$, let N_k be a uniform upper bound for the number of vertices in a radius- k ball of the base graph.

Lemma 5.2. *Fix $\sigma, \beta > 0$, and choose a real polynomial p of degree at most $k \geq 1$ such that*

$$\sup_{x \in [-1, 1]} |p(x) - (x - (Kb + \sigma))_+^2| \leq \beta.$$

Suppose a base partial assignment, with unassigned set $\mathcal{U}$, and a slab length $L \geq \max\{1, 4k \log J\}$ have surviving cost

$$\Psi = \frac{1}{L} \int_{V \times [0, L]} m_u^{\text{surv}} d\tilde{\mu}(u),$$

where m^{surv} is defined from (5.1). Complete the base coloring by giving $\mathcal{U}$ one additional color, and denote its paved matrix system by T_{pav} . Then

$$\int_V ((T_{\text{pav}} - (Kb + \sigma)\mathbf{1})_+^2)_{vv} d\mu(v) \leq 2\beta + \mu(\mathcal{U}) + \frac{2J^k N_k}{\sigma} \Psi. \quad (5.13)$$

Proof. Put $g(x) = (x - (Kb + \sigma))_+^2$. Since the paved operator is a compression by mutually orthogonal color projections, it remains a self-adjoint contraction; in particular $0 \leq g(T_{\text{pav}}) \leq \mathbf{1}$.

For an assigned base vertex v , let $B_k(v)$ be its radius- k ball in the base graph, and let T_v be the finite compression of the paved matrix to the assigned vertices of this ball. Polynomial diagonal moments through degree k at v are unchanged by this compression. Indeed every walk contributing to such a moment stays within $B_k(v)$, and edges from assigned to unassigned vertices are zero after paving. Consequently

$$(p(T_{\text{pav}}))_{vv} = (p(T_v))_{vv}.$$

Let $\mathcal{F}$ be the measurable set of assigned vertices for which $\lambda_{\max}(T_v) > Kb + \sigma$. Measurability follows by finite matrix evaluation on the Borel balls. If $v \notin \mathcal{F}$ is assigned, then $g(T_v) = 0$ and the two polynomial approximation errors give

$$(g(T_{\text{pav}}))_{vv} \leq 2\beta.$$

On $\mathcal{F} \cup \mathcal{U}$ its diagonal is at most 1. Therefore

$$\int_V (g(T_{\text{pav}}))_{vv} d\mu(v) \leq 2\beta + \mu(\mathcal{U}) + \mu(\mathcal{F}). \quad (5.14)$$

It remains to bound $\mu(\mathcal{F})$. Set $I_{\text{int}} = [k \log J, L - k \log J]$. For $u = (v, s)$ with $s \in I_{\text{int}}$, the entire ball $B_k(v)$ lifts into the slab: the height change along each graph edge has absolute value at most $\log J$. Let Q_u project onto the assigned vertices of this lifted ball. With A_L as in (5.1), the order inequality $A_L \leq (A_L)_+$ gives, after compression and diagonal congruence,

$$T_v - Kb\mathbf{1} \leq d_{\text{ball}}^{-1/2} Q_u (A_L)_+ Q_u d_{\text{ball}}^{-1/2}. \quad (5.15)$$

The base and lifted finite matrices here are identified up to the allowed diagonal unitary gauge, which does not affect the inequality or any diagonal spectral quantities. The right side is positive. If $v \in \mathcal{F}$, its norm is greater than σ by (5.15), and hence its trace is greater than σ . Equivalently,

$$\sum_{y \in B_k(u) \text{ assigned}} \frac{m_y^{\text{surv}}}{d_y} > \sigma. \quad (5.16)$$

Here $B_k(u)$ denotes the lift of the base ball, which agrees with the radius- k enclosing lifted ball at these interior heights. The comparison uses only $A_L \leq (A_L)_+$; it does not require positive part to commute with compression or diagonal congruence.

Integrate (5.16) against $d\mu(v) ds/L$ over $\mathcal{F} \times I_{\text{int}}$, and enlarge the resulting nonnegative sum to all interior roots. Since $d\mu(v) ds = d_u d\tilde{\mu}(u)$, the resulting integrand is a sum of $(d_u/d_y) m_y^{\text{surv}}$. On a radius- k ball, $d_u/d_y \leq J^k$. Exchanging roots and targets by (4.5), each target y has at most N_k possible roots. Deleted vertices may be retained as locations for the enclosing-ball relation, with $m_y^{\text{surv}} = 0$ there. Thus

$$\sigma \frac{|I_{\text{int}}|}{L} \mu(\mathcal{F}) \leq J^k N_k \Psi.$$

Our lower bound on L ensures $|I_{\text{int}}|/L \geq 1/2$, including the case $J = 1$. We conclude that

$$\mu(\mathcal{F}) \leq \frac{2J^k N_k}{\sigma} \Psi.$$

Combining this with (5.14) proves (5.13). $\square$

Completion of the proof of Theorem 4.1. Fix the prescribed $\sigma, \rho > 0$. Choose $\beta > 0$ with $2\beta < \rho/4$, and then choose the polynomial and its degree $k \geq 1$ in Lemma 5.2. Thus N_k is fixed before selecting the slab. Take

$$\eta = \rho/4, \quad \zeta = \frac{\rho\sigma}{8J^k N_k}, \quad L_0 = \max\{1, 4k \log J\}.$$

Proposition 5.1 supplies a partial assignment and a slab of length at least L_0 with $\mu(\mathcal{U}) < \eta$ and $\Psi < \zeta$. Give $\mathcal{U}$ the additional color and apply Lemma 5.2. Its right side is strictly less than ρ , as required.

Only the r assigned colors and one extra color occur in the final partition. The auxiliary graph colors, the slab length, the approximation degrees and the number of updates affect its construction but not this number of final colors. In particular the graph and cocycle bounds do not enter the condition on r in the theorem. $\square$

6 Expected maximal abelian subalgebras

We now combine the measured paving theorem with a separate argument for the part of an operator orthogonal to the normalizer algebra. Throughout this section, M has separable predual, φ is a faithful normal state, and $A \subset M$ is a MASA contained in the centralizer M_φ . We write E_A for the φ -preserving normal conditional expectation and use the sequence quotient of Lemma 2.2. Brackets denote classes in that quotient; a fixed operator also denotes its constant sequence. A partition in the quotient will always be represented by partitions in A with one fixed finite number of members.

Theorem 6.1. *For every $\gamma > 0$ there is an integer $R_{\text{st}}(\gamma)$ with the following property. For every self-adjoint $x \in M$ with $\|x\| \leq 1$, there are partitions $\mathcal{P}_j$ of 1 in A , each with $R_{\text{st}}(\gamma)$ members, and self-adjoint $w_j \in M$ such that*

$$\|w_j\| \leq 2, \quad \|w_j\|_\varphi \rightarrow 0, \quad \left\| C_{\mathcal{P}_j}(x - w_j) - E_A(x) \right\| \leq \gamma. \quad (6.1)$$

The integer $R_{\text{st}}(\gamma)$ is independent of M, A, φ , and x .

6.1 The normalizer algebra

Let N be the von Neumann algebra generated by the groupoid normalizers of A in M . These are the partial isometries v with $v^*v, vv^* \in A$ such that conjugation by v carries Av^*v onto Avv^* . The algebra N contains A . Since the modular group of φ fixes A pointwise, it carries every groupoid normalizer to another with the same supports and the same induced corner isomorphism. Hence N is globally modular invariant. Takesaki's theorem [Takesaki72] gives a normal φ -preserving conditional expectation $E_N : M \rightarrow N$; uniqueness of the state-preserving expectation onto A gives

$$E_A E_N = E_A. \quad (6.2)$$

The inclusion $A \subset N$ is Cartan. For completeness, groupoid regularity here implies regularity by unitary normalizers. Represent A on a standard probability space. A normal isomorphism between

two corners of A is represented by a nonsingular measurable partial bijection. On its fixed-point set, the corresponding partial normalizer belongs to A . On the complement, a countable separating family of measurable sets partitions the domain into countably many pieces on each of which the domain and range are disjoint. If v is the restriction to one such piece, then

$$v + v^* + 1 - v^*v - vv^*$$

is a unitary normalizer. Multiplication by its source projection recovers v , and strong summation recovers the original partial normalizer. Also A is a MASA in N and $E_A|_N$ is a faithful normal expectation, as required for the Cartan theorem.

The Feldman–Moore representation [FM77] identifies A with $L^\infty(V, \mu)$, where μ is the probability measure given by $\varphi|_A$, and identifies N with the algebra of a countable nonsingular equivalence relation, allowing a scalar unitary cocycle. We use the representation on square-integrable columns, with the right base point integrated against μ and counting measure in its orbit. An algebra element acts by left matrix multiplication on the orbit column. In the twisted case, changing the base point changes the matrix by diagonal unitary conjugacy, by the scalar cocycle identity. The conditional expectation onto A is its rooted diagonal. This is $E_A|_N$: if $F : N \rightarrow A$ is any normal conditional expectation, bimodularity gives $F(C_Q(z)) = F(z)$. The bounded compression net converges ultraweakly to $E_A(z)$ by Lemma 2.1, so normality gives $F(z) = F(E_A(z)) = E_A(z)$.

Bounded graph partial-isometry kernels and diagonal coefficients generate N . Their algebraic span, including products, is strongly dense. A finite such expression has off-diagonal support in a finite union of partial-bijection graphs, and therefore in an undirected graph of uniformly bounded degree. Operator-norm bounds in the column representation give the same matrix bounds for almost every orbit. All models and cocycle identities below may be restricted to a common conull invariant set for the countably many operators involved.

Fix $x = x^*$ with $\|x\| \leq 1$, and put

$$a_0 = E_A x, \quad Y = E_N x - a_0, \quad Z = x - E_N x. \quad (6.3)$$

Then $x - a_0 = Y + Z$, with $\|Y\|, \|Z\| \leq 2$ and $E_N Z = 0$. The measured theorem will control Y through its relation matrices. For Z , a unitary dilation with a diffuse left–right kernel will provide random diagonal compressions whose moments approach a free compression law. Its norm bound will make the spectral excess state-null, giving a second quotient norm estimate. Common refinement combines the two bounds; clipping the resulting block-diagonal residual then supplies the correction in Theorem 6.1.

Lemma 6.2. *Let $b, \sigma > 0$ and let r be an integer such that*

$$2\sqrt{2/r} + 2/r < b.$$

There are sequences of A -partitions with $(r+1)^2$ members whose quotient partition $\mathcal{P}$ satisfies

$$\|C_{\mathcal{P}}[Y]\| \leq 6(Kb + \sigma), \quad (6.4)$$

where K is the universal constant in Theorem 4.1.

Proof. Bounded Kaplansky approximation, followed by norm approximation inside the generating algebra, gives self-adjoint finite graph expressions converging strongly-star to Y , with norms at most 3. Subtracting their A -expectations makes their diagonals zero and increases the norm bound to 6; state-norm contractivity of E_A preserves convergence. For each such expression, cut down by the

base set on which every incident edge in its finite-degree enclosing graph has cocycle ratio between J^{-1} and J . These sets increase to full measure as $J \rightarrow \infty$, and the cuts converge strongly-star to the expression. A diagonal choice therefore gives self-adjoint Y_j with

$$E_A Y_j = 0, \quad \|Y_j\| \leq 6, \quad \|Y_j - Y\|_\varphi^\# \rightarrow 0,$$

whose orbit matrices have bounded-degree support and bounded cocycle ratios, with the bounds allowed to depend on j .

Apply Theorem 4.1 to $Y_j/6$ and separately to $-Y_j/6$, using state-error tolerances tending to zero. The rooted diagonal formula for the state says that the positive part above $Kb + \sigma$ has φ -norm tending to zero in each case. These positive parts are self-adjoint and uniformly bounded, so they belong to the null ideal of Lemma 2.2. The sequence $Y_j - Y$ belongs to that ideal as well. The resulting two quotient partitions, each of size $r + 1$, therefore give respectively

$$C_{\mathcal{P}^+}[Y/6] \leq (Kb + \sigma)1, \quad C_{\mathcal{P}^-}[Y/6] \geq -(Kb + \sigma)1.$$

Take their common refinement. All partition projections commute, and compression is positive and unital, so both scalar bounds hold after refinement. This proves the lemma. Further finite A -refinement preserves the bound. $\square$

We have completed the normalizer part. The remaining component has the following target, whose proof is given after the kernel and free-moment lemmas below.

Proposition 6.3. *For every $n \geq 2$, there are sequences of n -partitions in A whose quotient partition $\mathcal{P}$ satisfies*

$$\|C_{\mathcal{P}}[Z]\| \leq \frac{4}{\sqrt{n}}. \quad (6.5)$$

6.2 The diffuse kernel outside the normalizer algebra

We retain the faithful normal state φ , with $A \subset M_\varphi$, and the algebra N generated by the groupoid normalizers of A . Use a standard probability model $A = L^\infty(V, \mu)$ for $\varphi|_A$. We use the standard-form realization $L^2(M)$, equivalently Haagerup $L^2(M)$, and write $\varphi^{1/2}$ for the positive vector implementing φ . In particular, $a\varphi^{1/2} = \varphi^{1/2}a$ for $a \in A$.

For $X \in M$, the commuting left and right actions of A give a finite joint spectral measure ν_X for $X\varphi^{1/2}$ on $V \times V$. Our first coordinate is the right coordinate: for Borel sets $B, C \subset V$,

$$\nu_X(B \times C) = \|1_C X \varphi^{1/2} 1_B\|_2^2 = \int_B E_A(X^* 1_C X)(v) d\mu(v). \quad (6.6)$$

Both marginal measures are absolutely continuous with respect to μ . The right marginal has density $E_A(X^* X) \leq \|X\|^2$; the left marginal is absolutely continuous because $a \mapsto \varphi(X^* a X)$ is normal on A . Disintegration on standard Borel spaces therefore gives a measurable kernel $k_X(v, dw)$ such that

$$\nu_X(dv, dw) = d\mu(v) k_X(v, dw), \quad E_A(X^* a X)(v) = \int_V a(w) k_X(v, dw), \quad k_X(v, V) \leq \|X\|^2 \quad (6.7)$$

almost everywhere, for $a \in A$. As usual, individual function identities are understood modulo null sets. The joint spectral resolution can be constructed from a bounded real coordinate generating the standard model and the commuting left and right spectral resolutions of that coordinate.

Lemma 6.4. *If $E_N(X) = 0$, then $k_X(v, \cdot)$ has no atoms for almost every $v \in V$.*

Proof. Choose Borel representatives for the kernel. The set

$$\mathcal{T} = \{(v, w) : k_X(v, \{w\}) > 0\}$$

is Borel and has countable sections in the second coordinate. For example, joint measurability of the singleton mass follows by using shrinking neighborhoods in a compatible separable metric on the standard Borel space. The countable-section uniformization theorem expresses $\mathcal{T}$ as a countable union of graphs of partial Borel maps. If the conclusion fails, one such map $F : S \rightarrow V$ has a domain of positive μ -measure, with

$$g(v) := k_X(v, \{F(v)\}) > 0 \quad (v \in S).$$

There is no assumption that F is injective. For a μ -null Borel set $B \subset V$, absolute continuity of the left marginal gives

$$\int_{S \cap F^{-1}(B)} g(v) d\mu(v) \leq \nu_X(V \times B) = 0.$$

Thus F pulls back null sets to null sets. With $e = 1_S \in A$, composition with F consequently defines a normal $*$ -homomorphism

$$\theta : A \longrightarrow Ae, \quad \theta(a)(v) = a(F(v)) \quad (v \in S), \quad \theta(1) = e.$$

The graph will intertwine A with the possibly proper subalgebra $\theta(A)$ of Ae ; its polar partial isometry need not yet normalize A . We will use an expected type I intermediate algebra to split the right support into corners where these two abelian algebras coincide.

Apply the joint spectral projection of the graph of F to $X\varphi^{1/2}$ and call the resulting vector ζ . It is nonzero and satisfies

$$\zeta e = \zeta, \quad a\zeta = \zeta\theta(a) \quad (a \in A). \quad (6.8)$$

Its right spectral density on S is g . In particular, e is its smallest right supporting projection among the projections of A : if $f \in A$ is a projection and $\zeta f = \zeta$, then $e \leq f$.

Take the L^2 polar decomposition $\zeta = uh$, where $h = |\zeta| \in L^2(M)_+$ and $u \in M$ is a partial isometry. Put $p = u^*u \leq e$ and $q = uu^*$. We use here the usual polar decomposition and support properties of standard-form noncommutative L^2 ; see, for example, [Hiai20, Lemma 11.12 and Theorem 11.30]. For a unitary $a \in A$, the operator $b = \theta(a) + 1 - e$ is unitary and (6.8) reads $a\zeta = \zeta b$. Taking absolute values gives $h = b^*hb$. Uniqueness of the polar decomposition then gives $au = ub = u\theta(a)$. Consequently,

$$h\theta(a) = \theta(a)h, \quad p\theta(a) = \theta(a)p, \quad au = u\theta(a) \quad (a \in A), \quad q \in A. \quad (6.9)$$

The identities for all a follow by linearity from unitaries. For the last assertion, the identities imply that q commutes with A , and A is maximal abelian.

Consider the von Neumann algebra

$$D = \theta(A)' \cap eMe.$$

It contains p and Ae , and Ae is a MASA in D . Moreover $\theta(A) \subset Z(D)$. If $d \in pDp$, then udu^* commutes with Aq , by (6.9), and hence belongs to Aq . On the other hand,

$$u\theta(a)pu^* = aq \quad (a \in A).$$

Conjugation by u therefore identifies pDp with Aq , and

$$pDp = \theta(A)p. \quad (6.10)$$

Let c be the central support of p in D . Since $Z(D) \subset Ae$, the identity $pc = p$ gives $\zeta c = \zeta$; the minimal right A -support property gives $c = e$. Thus D has an abelian projection with full central support and is type I. Finally, if $d \in Z(D)$, then (6.10) gives $dp = \theta(a)p$ for some $a \in A$. The central element $d - \theta(a)$ annihilates a projection of full central support and is zero. We have proved

$$Z(D) = \theta(A). \quad (6.11)$$

We next need a partition of e by projections $e_j \in Ae$ such that

$$(Ae)e_j = \theta(A)e_j. \quad (6.12)$$

The restriction of E_A to D is a normal conditional expectation onto Ae . The type I MASA theorem of Akemann and Sherman [AS12, Theorem 4.1] supplies a partition of e by projections in Ae that are abelian in D . There are only countably many nonzero members, because D has the faithful normal state $\varphi|_D/\varphi(e)$. For an abelian projection in a type I algebra,

$$e_j D e_j = Z(D) e_j.$$

This follows, for instance, by writing the type I algebra as a direct integral: an abelian projection has fiber rank at most one. Together with (6.11) and $\theta(A) \subset Ae \subset D$, this proves (6.12).

Set $\eta_j = \zeta e_j$ and take its polar decomposition $\eta_j = u_j h_j$. Its right support is at most e_j . Repeating (6.9) shows that h_j commutes with $\theta(A)$ and that

$$au_j = u_j \theta(a) \quad (a \in A).$$

By (6.12) and the support condition, h_j commutes with all of A . Therefore the right support $p_j = u_j^* u_j$ belongs to A ; the left support $q_j = u_j u_j^*$ belongs to A by the same polar argument as before. In addition,

$$\theta(A)p_j = Ap_j, \quad u_j(Ap_j)u_j^* = Aq_j.$$

Thus each nonzero u_j is a groupoid normalizer and belongs to N .

Let ψ_j be the normal positive functional whose positive implementing vector is h_j . Since h_j commutes with A , this functional centralizes A . In particular $\psi_j \circ C_Q = \psi_j$ for every finite partition Q in A . Lemma 2.1, normality, and boundedness of the compression net imply

$$\psi_j = \psi_j \circ E_A.$$

Write $f_j = d(\psi_j|_A)/d\mu \in L^1(V, \mu)_+$ and $f_{j,m} = \min(f_j, m)$. The positive vectors $f_{j,m}^{1/2} \varphi^{1/2}$ belong to $A\varphi^{1/2}$ and converge in L^2 ; indeed their squared pairwise distances are

$$\int_V |f_{j,m}^{1/2} - f_{j,n}^{1/2}|^2 d\mu.$$

Their normal positive functionals are $x \mapsto \int f_{j,m} E_A(x) d\mu$ and converge in norm to ψ_j . The limit is therefore the unique positive implementing vector h_j . It follows that

$$h_j \in \overline{A\varphi^{1/2}}, \quad \eta_j \in \mathcal{H}_N := \overline{N\varphi^{1/2}}.$$

The right projections e_j are orthogonal and sum to e , so $\zeta = \sum_j \zeta e_j \in \mathcal{H}_N$, with convergence in L^2 .

The closed space $\mathcal{H}_N$ reduces both the left and right actions of A : for the right action, use $n\varphi^{1/2}a = na\varphi^{1/2}$. It therefore reduces their joint spectral projections, including the graph projection defining ζ . But $E_N(X) = 0$ gives, for $n \in N$,

$$\varphi(n^*X) = \varphi(n^*E_N(X)) = 0,$$

so $X\varphi^{1/2} \perp \mathcal{H}_N$. Thus $\zeta \perp \mathcal{H}_N$, contradicting $\zeta \neq 0$ and the membership just proved. $\square$

Removing atomic summands. For the remaining treatment of $Z = x - E_N(x)$ we may assume that A is diffuse. Here are the details of this reduction. If f is an atom of A , then $Af = \mathbb{C}f$ is a MASA in fMf , so $fMf = \mathbb{C}f$. Thus f is minimal in M . Let z_f be its central support in M . The algebra z_fM is type I, since f is an abelian projection with full central support there. Compression by f is faithful on its center and has scalar range, so z_fM is a type I factor.

The algebra Az_f is a MASA in this factor and has the normal expectation $E_A|_{z_fM}$. Again by [AS12, Theorem 4.1], it has a partition of z_f into rank-one projections. In particular it is purely atomic. Partial isometries joining any two of these rank-one projections normalize their A -corners. A compatible system of these matrix units generates z_fM , and hence

$$z_fM \subset N.$$

Distinct central supports of atoms are either equal or orthogonal, because each supports a factor summand. Their sum z_{at} is a central projection, belongs to A , and satisfies $z_{\text{at}}M \subset N$. Consequently

$$z_{\text{at}}Z = 0.$$

On the complementary central summand, $A(1 - z_{\text{at}})$ has no atoms. The algebra generated there by its groupoid normalizers is $N(1 - z_{\text{at}})$: corner normalizers extend by zero, and global normalizers restrict to this central summand. We may use the normalized restriction of φ on the nonzero complementary summand. This changes state seminorms only by a fixed scalar factor. A partition obtained there extends to M with the same number of cells by adding z_{at} to one cell; all estimates for Z are preserved because Z vanishes on z_{at} .

6.3 Random diagonal unitaries and free moments

It remains to compress the diffuse summand of Z . The next two lemmas supply the free law and its norm bound for the matrix dilation used to prove Proposition 6.3. The state in the probabilistic argument need not be tracial.

Lemma 6.5. *Let $\mathcal{M}$ have separable predual, let ϕ_0 be a faithful normal state, and let $D \subset \mathcal{M}_{\phi_0}$ be a diffuse MASA. Suppose that $U \in \mathcal{M}$ is a self-adjoint unitary with $\phi_0(U) = 0$, and that the normal positive map*

$$\mathcal{K}(a) = E_D(UaU), \quad a \in D,$$

has an almost everywhere diffuse kernel on a standard probability model of $(D, \phi_0|_D)$. For every integer $n \geq 2$ there are unitaries $w_l \in D$, with $w_l^n = 1$, such that the joint star moments of (w_l, U) converge to those of free unitaries (w, U_f) , where w is uniform on the n th roots of unity and U_f is uniform on $\{-1, 1\}$.

Proof. Choose an increasing sequence of finite partitions $\mathcal{Q}$ in D which generates D , separates points of a conull standard model, and satisfies

$$\max_{q \in \mathcal{Q}} \phi_0(q) \longrightarrow 0. \tag{6.13}$$

Such a sequence is obtained by adjoining a countable separating family and, at each stage, a sufficiently small-mass finite refinement; the latter exists by diffuseness. Give its cells independent labels Z_q uniform on the n th roots of unity, and put $w_{\mathcal{Q}} = \sum_q Z_q q$.

We first make the averaged reduced moments vanish by estimating return probabilities for diffuse kernels. We then control their fluctuations using the small masses of the partition cells, so that one deterministic labeling works for each prescribed finite list of moments.

Averaged words and a repeated-cell compression. We first prove that the label expectation tends to zero for each scalar

$$F_Q = \phi_0(x_0 w_Q^{k_1} x_1 \cdots w_Q^{k_h} x_h), \quad h \geq 1, \quad 1 \leq k_j < n, \quad (6.14)$$

where $x_j = U$ for $1 \leq j < h$ and each endpoint x_0, x_h is either 1 or U . Expand over cell assignments $q_1, \dots, q_h$. An assignment survives label averaging only if, in every block of equal cells, the sum of its exponents is zero modulo n . Consequently every block of its exact equality pattern has size at least two. For any surviving exact pattern, inclusion-exclusion expresses its word sum as a finite linear combination of sums

$$S_\pi = \sum_{\substack{q_1, \dots, q_h \in Q \\ q_i = q_j \text{ if } i, j \text{ are in one block of } \pi}} \phi_0(x_0 q_1 x_1 \cdots q_h x_h), \quad (6.15)$$

where every block of π has at least two members. There is no distinctness requirement between different blocks in (6.15). It is enough to show $S_\pi \rightarrow 0$ for each such π .

Choose positions $c < d$ in one block whose distance $d - c$ is minimal among all pairs in a common block. The positions strictly between them belong to distinct other blocks. We impose the equalities of π using auxiliary Haar phases. For a block other than the selected block, place independently sampled cellwise Haar diagonal unitaries at all but one of its occurrences, and their inverse product at the remaining occurrence. Haar averaging forces precisely the equality of the cells at those occurrences. In the selected block, use the same projection q explicitly at c and d and sum over q ; put independent Haar diagonals at every other occurrence of that block, with their inverse product additionally at c . All initial draws are independent. Expanding and averaging now gives exactly S_π as the auxiliary expectation of

$$\phi_0(\text{PRE } C_Q(Y_Q) \text{ POST}), \quad Y_Q = U d_{c+1} U \cdots d_{d-1} U. \quad (6.16)$$

When $d = c + 1$, the last expression is U . The extra inverse product at c can be moved before its projection, since both lie in D , and is included in PRE. The factors outside the compression are products of fixed contractions x_j and diagonal unitaries. Each interior d_j is a cellwise Haar diagonal: this is also true if it is the inverse product placed at the last occurrence of its block. The interior diagonals are mutually independent because their positions belong to different blocks. Independence from PRE and POST is not asserted or needed.

Their independence and orthogonality of the outside projections give

$$\mathbb{E} \|C_Q(Y_Q)\|_{\phi_0}^2 = \sum_{q, q_{c+1}, \dots, q_{d-1} \in Q} \|q U q_{c+1} U \cdots q_{d-1} U q\|_{\phi_0}^2. \quad (6.17)$$

There are no middle cells when $d = c + 1$. Fix a coarser partition $\mathcal{E}$ from the generating sequence. When Q refines $\mathcal{E}$, contractivity of C_Q in the state norm, applied with each middle cell tuple fixed, bounds the right side by

$$\sum_{p \in \mathcal{E}} \sum_{q_{c+1}, \dots, q_{d-1} \in Q} \|p U q_{c+1} U \cdots q_{d-1} U p\|_{\phi_0}^2. \quad (6.18)$$

Diffuse kernel return probabilities. Here is the limit of this expression without a tracial assumption. Put $m = d - c$. For each $p \in \mathcal{E}$ define

$$B_{1,Q}(p) = U p U, \quad B_{j+1,Q}(p) = U C_Q(B_{j,Q}(p)) U.$$

Expanding the squares in (6.18) and summing the middle cells identifies that expression with $\sum_{p \in \mathcal{E}} \phi_0(pB_{m,\mathcal{Q}}(p)p)$. All $B_{j,\mathcal{Q}}(p)$ are positive contractions. By Lemma 2.1, $C_{\mathcal{Q}}(t) \rightarrow E_D(t)$ strongly-star on each fixed bounded element. Its contractivity for both state seminorms also controls a bounded, varying, strong-star null error. Bounded strong-star continuity of multiplication by the fixed U therefore proves recursively that $B_{j,\mathcal{Q}}(p)$ converges strongly-star to $B_j(p)$, where

$$B_1(p) = UpU, \quad B_{j+1}(p) = UE_D(B_j(p))U.$$

Thus $E_D(B_j(p)) = \mathcal{K}^j(p)$. If μ_0 denotes the probability measure on D , state preservation and bimodularity of E_D show that the limit of (6.18) is

$$\sum_{p \in \mathcal{E}} \phi_0(p\mathcal{K}^m(p)p) = \sum_{p \in \mathcal{E}} \int_p (\mathcal{K}^m 1_p)(v) d\mu_0(v). \quad (6.19)$$

The positive normal map $\mathcal{K}$ is unital. Its kernel is nonsingular in the integrated sense: a fixed μ_0 -null set receives zero mass for almost every starting point. This follows from normality, since its indicator is zero in D . In particular the kernel avoids, for almost every starting point, the fixed exceptional null set on which its successor kernel is not diffuse. Integrating diffuse successor kernels consequently produces a diffuse kernel. Induction shows that each fixed iterate $\mathcal{K}^m$ has an almost everywhere diffuse kernel. As $\mathcal{E}$ increases along the separating sequence, the cell containing v decreases to $\{v\}$ on the chosen conull model. Its return mass in (6.19) therefore tends to zero for almost every v . It is bounded by $\mathcal{K}^m 1(v) = 1$. Dominated convergence proves that (6.19) tends to zero. First fixing $\mathcal{E}$ and letting $\mathcal{Q}$ refine, and then letting $\mathcal{E}$ refine, proves

$$\mathbb{E} \|C_{\mathcal{Q}}(Y_{\mathcal{Q}})\|_{\phi_0}^2 \rightarrow 0. \quad (6.20)$$

Nonracial state estimates. We next explain the state estimates that turn this into scalar moment convergence. For an entire analytic element a for the modular group, the standard-form right-action identity gives

$$\|ta\|_{\phi_0} \leq \left\| \sigma_{i/2}^{\phi_0}(a) \right\| \|t\|_{\phi_0}. \quad (6.21)$$

See [Hiai20, Lemma 8.12, Equation (8.4)], applied to a^* . Entire analytic elements approximate each fixed element boundedly in the state norm, for example by Gaussian modular smoothing [Hiai20, Lemma 2.13]. A centralizer unitary has right-multiplication bound 1. In any scalar word consisting of fixed bounded factors and varying centralizer unitaries, replace the fixed factors from right to left by bounded analytic approximants. At a replacement step all factors to the right already have uniformly bounded right actions, and all factors to the left have uniformly bounded operator norms. The error in the scalar state value is therefore at most a fixed constant times the state-norm error of the factor being replaced. Choosing the approximants in that order makes the total error arbitrarily small, uniformly over the varying diagonals and partitions.

Apply this procedure only to the fixed factors in POST in (6.16). The compression and all remaining left factors are contractions. With the approximants fixed, the absolute scalar value is bounded by $C \|C_{\mathcal{Q}}(Y_{\mathcal{Q}})\|_{\phi_0}$, where C is independent of every phase and of $\mathcal{Q}$. Cauchy–Schwarz in the auxiliary probability space and (6.20) make its phase expectation tend to zero. The uniformly arbitrarily small replacement error proves $S_{\pi} \rightarrow 0$. Hence

$$\mathbb{E} F_{\mathcal{Q}} \rightarrow 0. \quad (6.22)$$

Concentration and deterministic labels. It remains to choose deterministic labels simultaneously for the moments. Replace all fixed factors in (6.14) analytically, again from right to left, and call the resulting scalar G_Q . It approximates F_Q uniformly to any prescribed accuracy. Changing one label Z_q changes a power of w_Q by a scalar of absolute value at most 2 times q . Telescoping over the finitely many occurrences, each difference is thus a bounded scalar times $\phi_0(\text{PRE } q \text{ POST})$. Cauchy–Schwarz gives

$$\begin{aligned} |\phi_0(\text{PRE } q \text{ POST})| &\leq \|q\text{PRE}^*\|_{\phi_0} \|q\text{POST}\|_{\phi_0} \\ &\leq C_{\text{word}} \phi_0(q). \end{aligned} \tag{6.23}$$

The second line follows from (6.21) for both products, including their analytic adjoints. Their modular norm bounds are uniform over all other labels, since the diagonal unitaries are in the centralizer. Independent bounded differences (or the martingale variance sum obtained by revealing one label at a time) now give

$$\mathbb{E} |G_Q - \mathbb{E} G_Q|^2 \leq C'_{\text{word}} \sum_{q \in Q} \phi_0(q)^2 \rightarrow 0$$

by (6.13). The uniform approximation shows that $F_Q - \mathbb{E} F_Q \rightarrow 0$ in probability. Together with (6.22), this gives $F_Q \rightarrow 0$ in probability.

There are only countably many test words. For successively larger finite lists, convergence in probability gives a partition far enough along the sequence and a deterministic labeling for which every word in the list is as small as prescribed. A diagonal choice gives w_l with all those moments tending to zero. The tests include the nontrivial single powers of w_l , so its limiting distribution is uniform on the n th roots. Also $U^2 = 1$ and $\phi_0(U) = 0$ give the symmetric two-point distribution. Centered elements of the two limiting generated algebras are linear combinations of the tested powers and of U , respectively; all alternating centered moments are precisely the tests above. This establishes the asserted joint free law. $\square$

Lemma 6.6. *Let w and U_f have the free law in Lemma 6.5, in its reduced free-product representation. If $p_1, \dots, p_n$ are the spectral projections of w , then*

$$\left\| \sum_{j=1}^n p_j U_f p_j \right\| \leq \frac{2}{\sqrt{n}}. \tag{6.24}$$

Proof. Let E_0 be the Hilbert-space projection onto the vacuum and all reduced words not starting in the centered part of the w -algebra. For a spectral projection p of w , put $t = 1/n$. On this subspace, left multiplication by p is the sum of the scalar action t and creation by $p - t1$. The two resulting subspaces are orthogonal, and $\|p - t1\|_2^2 = t(1 - t)$; hence

$$\|pE_0\| = \|E_0p\| = \sqrt{t}.$$

Since U_f is centered, it sends a word starting in the w -centered space to a word starting in the other factor. Therefore $U_f(1 - E_0) = E_0U_f(1 - E_0)$, and

$$pU_fp = pU_fE_0p + pE_0U_f(1 - E_0)p.$$

Unitarity gives norm at most $2\sqrt{t}$. The sum of the compressions is block diagonal for the mutually orthogonal p_j , so its norm is the largest block norm. $\square$

6.4 A free-product matrix dilation

Proof of Proposition 6.3. By the atomic reduction following Lemma 6.4, it is enough to work on the central summand where A is diffuse. Normalize the state on that summand and retain the same notation. Put $z = Z/2$, so z is a self-adjoint contraction and $E_N z = 0$.

The ordinary two-by-two unitary dilation need not have a diffuse kernel over $A \oplus A$: already at $z = 0$, its identity off-diagonal entry gives an atomic cross-kernel. We replace the second diagonal algebra by a free diffuse abelian algebra, turning the cross terms into normal functionals on diffuse spaces. This is the symmetry form of the free-product projection dilation in [PV15, proof of Theorem 5.1]: with $y = (1 + z)/2$, the symmetry below is twice that projection minus the identity. The kernel and moment lemmas above supply the required freeness in the present nonracial setting.

Let (B_0, τ_B) be a diffuse standard separable abelian probability algebra and form the reduced von Neumann free product

$$(P, \psi) = (M, \varphi) * (B_0, \tau_B).$$

The free-product state is faithful and normal, the two factor expectations preserve it, and its Hilbert-space representation is the reduced-word representation; see, for example, [HU16]. Both A and B_0 remain MASAs in P . We give the argument to make its nonracial applicability explicit. There are unitaries $v_k \in A$ tending weakly to zero, since A is diffuse. They centralize ψ , by the expectation onto M . On $L^2(P, \psi) \ominus L^2(M, \varphi)$, a dense family consists of vectors of words

$$x_0 b_1 x_1 \cdots b_h x_h, \quad h \geq 1,$$

where $b_i \in B_0$ are centered, the internal $x_i \in M$ are centered, and the endpoints are arbitrary. To compute a matrix coefficient of conjugation by v_k , take another such word with initial endpoint y_0 , and split the middle product $y_0^* v_k x_0$ into its centered part and its scalar mean. The centered part gives a reduced word with zero expectation onto M . The scalar part is $\varphi(y_0^* v_k x_0)$ times an expression with a fixed norm bound and tends to zero. The additional right factor v_k^* in the conjugation does not change that norm bound. Conjugation is unitary on the state Hilbert space; density proves that it tends weakly to zero on the indicated complement. If an element of P commutes with A , its vector after subtracting its M -expectation lies in this complement and is fixed by all these conjugations. It must be zero. The element belongs to M , and then to A . Interchanging the factors proves the assertion for B_0 .

In $\widetilde{M} = \text{Mat}_2(P)$ use the state $\widetilde{\varphi} = \text{tr}_2 \otimes \psi$, with normalized matrix trace. The diagonal algebra $D = A \oplus B_0$ is a diffuse MASA in its centralizer. Define

$$U = \begin{pmatrix} z & s \\ s & -z \end{pmatrix}, \quad s = (1 - z^2)^{1/2}. \quad (6.25)$$

Since z commutes with s , this is a self-adjoint unitary. Furthermore $E_D(U) = 0$. For $a \in A$ and $b \in B_0$, direct free-product expectation gives

$$E_D(U(a \oplus b)U) = (E_A(zaz) + \tau_B(b)E_A(s^2)) \oplus (\varphi(sas)1 + \tau_B(b)\varphi(z^2)1). \quad (6.26)$$

In particular, $E_{B_0}(zbz) = \tau_B(b)\varphi(z^2)1$ because $\varphi(z) = 0$. Lemma 6.4 makes the first kernel $a \mapsto E_A(zaz)$ diffuse. The other kernels in (6.26) are diffuse because the base algebras are diffuse and the displayed functionals are normal. Thus the kernel of $a \mapsto E_D(UaU)$ is almost everywhere diffuse.

Apply Lemma 6.5 with $\phi_0 = \widetilde{\varphi}$. Let $P_{j,l} \in D$, $1 \leq j \leq n$, be the spectral projections of the resulting finite-order w_l , padded with zero projections when necessary, and set

$$S_l = \sum_{j=1}^n P_{j,l} U P_{j,l}.$$

Each projection is a polynomial in w_l and w_l^* . Consequently the scalar moments of the self-adjoint contractions S_l tend to the moments of $S_f = \sum_j p_j U_f p_j$ in the reduced free law. Let $c = 2/\sqrt{n}$. Lemma 6.6 gives $\|S_f\| \leq c$. Uniform polynomial approximation on $[-1, 1]$, applied to the continuous function $t \mapsto (|t| - c)_+^2$, gives

$$\|(|S_l| - c)_+\|_{\tilde{\varphi}} \longrightarrow 0. \quad (6.27)$$

This passage uses moments to obtain a state-norm estimate, not operator-norm convergence in $\widetilde{M}$.

The sequence (S_l) belongs to the multiplier algebra of Lemma 2.2: U is fixed and each $(P_{j,l})_l$ is a bounded centralizer sequence. Its excess in (6.27) is self-adjoint, so it belongs to the strong-star null ideal. Continuous functional calculus in the quotient gives $\|[S_l]\| \leq c$.

Compress by the fixed first matrix projection e_{11} . Write $P_{j,l} = p_{j,l} \oplus b_{j,l}$, so the $(p_{j,l})_{j=1}^n$ form a partition in A . The compressed sequence is

$$e_{11} S_l e_{11} = \sum_{j=1}^n p_{j,l} z p_{j,l}$$

in the first corner. Its quotient norm upstairs is at most c . Functional calculus there makes its absolute-value excess above c strong-star null. These compressed elements belong to M , and the state on their corner restricts to $\varphi/2$; hence the same excess is strong-star null for (M, φ) . The compressed sequence is also a multiplier sequence for M , since its partition projections centralize φ . Its original quotient norm is therefore at most c . Multiplying by 2 proves (6.5) on the diffuse summand. Extending the same colors over the central summand where $Z = 0$ proves the full assertion. $\square$

6.5 Completion of the state-preserving paving argument

Proof of Theorem 6.1. Choose

$$b = \sigma = \frac{\gamma}{24(K+1)},$$

choose r with $2\sqrt{2/r} + 2/r < b$, and choose $n \geq 2$ with $4/\sqrt{n} \leq \gamma/2$. Use Lemma 6.2 and Proposition 6.3, and take the common refinement of their partition sequences. It has $R_{\text{st}}(\gamma) = (r+1)^2 n$ members after padding. Finite refinement preserves each quotient bound, so for

$$T_j^0 = C_{\mathcal{P}_j}(x - E_A x)$$

we have $\|[T_j^0]\| \leq \gamma$, since the two contributions are at most $\gamma/4$ and $\gamma/2$.

Let $\text{clip}_{\gamma}(t) = \max(-\gamma, \min(t, \gamma))$ for real t , and define

$$w_j = T_j^0 - \text{clip}_{\gamma}(T_j^0).$$

The quotient norm bound and continuous functional calculus show that (w_j) has zero quotient image. It is self-adjoint, hence $\|w_j\|_{\varphi} \rightarrow 0$. Also $\|T_j^0\| \leq 2$, so $\|w_j\| \leq 2$. The operator T_j^0 , and therefore w_j , is block diagonal for $\mathcal{P}_j$. Since $E_A x$ belongs to A , this gives

$$C_{\mathcal{P}_j}(x - w_j) - E_A x = T_j^0 - w_j = \text{clip}_{\gamma}(T_j^0),$$

of norm at most γ . These are all the assertions of (6.1). $\square$

7 Passage to arbitrary von Neumann algebras

We first remove the separability assumption from the state-preserving result. We then treat the part of an inclusion on which no normal positive functional centralizes the MASA. The two arguments will be combined using corners by projections in the MASA; these projections need not be central in the ambient algebra. The support e of the centralizing functionals will carry an expected MASA, while its complement h has none of those functionals. Only after choosing the finite vector tests will we split $e = f + g$, with a faithful state on the f -corner and a residual projection g that annihilates the test vectors.

7.1 Separable reduction

Lemma 7.1. *The conclusion of Theorem 6.1 holds without the assumption of separable predual, with the same partition-size bound.*

Proof. Let φ be a faithful normal state on M , let $A \subset M$ be a MASA in its centralizer, and fix a self-adjoint contraction $x \in M$. Construct countable sets of generators as follows. Start with $1, x$. At each stage form the unital $\mathbb{Q}(i)$ -algebra generated by the current set and its adjoints. For every element t of this countable algebra, add $E_A(t)$ and the projections of finite A -partitions $\mathcal{Q}_{t,k}$ satisfying

$$\|C_{\mathcal{Q}_{t,k}}(t) - E_A(t)\|_{\varphi}^{\#} < 1/k, \quad k \geq 1.$$

These partitions exist by Lemma 2.1. Repeat this procedure countably many times, let B be the union of the resulting countable algebras, and set

$$M_1 = W^*(B), \quad A_1 = A \cap M_1.$$

Every element of M_1 is the limit of an operator-norm bounded sequence from B in $\|\cdot\|_{\varphi}^{\#}$. Indeed, Kaplansky density gives bounded strong-star approximants from the norm closure of B , and one can then approximate in norm by elements of B . On bounded sets the state sharp seminorm determines the strong-star topology, so sequences suffice here. Taking real and imaginary parts separately gives this assertion for arbitrary elements. Since E_A is contractive for the state sharp seminorm and in operator norm, it follows that $E_A(M_1) \subset A_1$.

Suppose $z \in M_1$ commutes with A_1 . Choose bounded $t_j \in B$ converging to z in the state sharp seminorm, and choose one of the stored partitions $\mathcal{Q}_j \subset A_1$ for each t_j so that its compression error is less than $1/j$. Since $C_{\mathcal{Q}_j}(z) = z$, contractivity gives

$$\|z - E_A(z)\|_{\varphi}^{\#} \leq 2 \|z - t_j\|_{\varphi}^{\#} + \|C_{\mathcal{Q}_j}(t_j) - E_A(t_j)\|_{\varphi}^{\#} \longrightarrow 0.$$

Thus $z = E_A(z) \in A_1$, proving that A_1 is a MASA in M_1 . It lies in the centralizer of $\varphi|_{M_1}$. The GNS representation of this restricted faithful normal state is faithful and normal, and its Hilbert space is separable because the countable set B applied to the cyclic vector is dense. Hence M_1 has separable predual. The restriction of E_A is its state-preserving expectation onto A_1 . Applying Theorem 6.1 in M_1 proves the assertion, including the unchanged bound on the partition size. $\square$

7.2 The corner without centralizing normal functionals

We use standard form in the next argument. Write $a\xi b$ for the commuting left and right actions, with the right action of b given by Jb^*J . Every normal positive functional θ has a unique representative

ξ_θ in the natural positive cone. This cone is selfdual and is preserved by $\xi \mapsto u\xi u^*$ for a unitary u . Moreover,

$$\|\xi_\theta - \xi_\psi\|^2 \leq \|\theta - \psi\|_{M_*}. \quad (7.1)$$

These standard-form facts hold for arbitrary von Neumann algebras; see [Hiai20, Theorem 3.12]. The inequality also appears in [Araki74, Theorem 4(8), Equation (5.10)]. For two functionals in a non- σ -finite algebra, it can also be applied in the corner supported by $\text{supp}(\theta + \psi)$, where their sum is faithful.

Lemma 7.2. *Let $D \subset P$ be a MASA such that no nonzero normal positive functional on P centralizes D . For every normal state θ on P , every integer $m \geq 2$, and every $\alpha > 0$, there are a positive contraction $s \in P$ and an m -partition $(p_i)_{i=1}^m$ in D such that*

$$\theta(s) > 1 - \alpha, \quad p_i s p_i \leq m^{-1} p_i \quad (1 \leq i \leq m). \quad (7.2)$$

Proof. In standard form, each finite partition $\mathcal{Q}$ in D defines an orthogonal projection

$$P_{\mathcal{Q}}\xi = \sum_{q \in \mathcal{Q}} q\xi q.$$

These projections decrease under refinement to the projection onto the D -central vectors. A nonzero central vector ζ would give a nonzero D -central normal positive functional $a \mapsto \langle a\zeta, \zeta \rangle$: conjugation by a unitary of D can be transferred to the commuting right action. The hypothesis therefore implies $P_{\mathcal{Q}} \rightarrow 0$ strongly along the net of finite partitions.

Choose independent random labels Z_q , uniform on the m th roots of unity, and put $u = \sum_{q \in \mathcal{Q}} Z_q q$. For $1 \leq k < m$, independence and $\mathbb{E}Z_q^k = 0$ give

$$\mathbb{E} \langle u^k \xi_\theta u^{-k}, \xi_\theta \rangle = \langle P_{\mathcal{Q}} \xi_\theta, \xi_\theta \rangle \rightarrow 0.$$

Each overlap is real and nonnegative, by the natural-cone properties. Consequently, for a sufficiently fine partition there is one choice of labels making all these overlaps arbitrarily small. The states

$$\theta_j = \theta \circ \text{Ad}(u^{-j}), \quad 0 \leq j < m,$$

have natural-cone vectors $\xi_j = u^j \xi_\theta u^{-j}$. Their pairwise overlaps are among those just considered. Equation (7.1) shows that $\|\theta_j - \theta_k\|_{P_*}$ is arbitrarily close to 2 for every $j \neq k$.

Fix a small $\eta > 0$. The positive support projection of $\theta_j - \theta_k$ attains half the norm of that Hermitian functional, since its value at $\mathbf{1}$ is zero. Thus the labels can be chosen so that there are projections $q_{jk} \in P$ with

$$\theta_j(q_{jk}) \geq 1 - \eta, \quad \theta_k(q_{jk}) \leq \eta \quad (j \neq k). \quad (7.3)$$

For $0 < t < 1$ put

$$a_j = \frac{1}{m-1} \sum_{k \neq j} q_{jk}, \quad b_j = 1_{[1-t, 1]}(a_j).$$

Spectral calculus and Equation (7.3) yield

$$\theta_j(b_j) \geq 1 - \frac{\eta}{t}, \quad \theta_k(b_j)^{1/2} \leq \sqrt{\eta} + \sqrt{(m-1)t} \quad (k \neq j). \quad (7.4)$$

For the first estimate, use $\theta_j(1 - a_j) \leq \eta$ and $1 - b_j \leq t^{-1}(1 - a_j)$. For the second, the positive-summand inequality $1 - q_{jk} \leq (m-1)(1 - a_j)$ gives

$$\|(1 - q_{jk})b_j\|^2 = \|b_j(1 - q_{jk})b_j\| \leq (m-1)t.$$

Splitting $b_j \xi_k = b_j q_{jk} \xi_k + b_j(1 - q_{jk}) \xi_k$ proves the claim, using the adjoint of this operator-norm estimate.

The pairwise separating projections need not be mutually orthogonal, so they cannot simply be added to form a partition. Sequential products give positive effects with sum at most the identity while retaining almost full mass for their respective states. Define

$$V_0 = \mathbf{1}, \quad V_j = (1 - b_{j-1}) \cdots (1 - b_0), \quad s_j = V_j^* b_j V_j \quad (0 \leq j < m).$$

These positive elements need not be projections. They satisfy

$$s_j = V_j^* V_j - V_{j+1}^* V_{j+1}, \quad \sum_{j=0}^{m-1} s_j = \mathbf{1} - V_m^* V_m \leq \mathbf{1}. \quad (7.5)$$

Telescoping the products of contractions also gives

$$\|V_j \xi_j - \xi_j\| \leq \sum_{k < j} \|b_k \xi_k\|.$$

By Equation (7.4), whenever $\eta < t$,

$$\|b_j V_j \xi_j\| \geq \sqrt{1 - \eta/t} - j(\sqrt{\eta} + \sqrt{(m-1)t}).$$

For fixed m , choosing t small and then η sufficiently small makes every $\theta_j(s_j) = \|b_j V_j \xi_j\|^2$ arbitrarily close to 1.

Set

$$s = \frac{1}{m} \sum_{j=0}^{m-1} u^{-j} s_j u^j.$$

Then $0 \leq s \leq \mathbf{1}$ and $\theta(s)$ is the average of the $\theta_j(s_j)$, so it exceeds $1 - \alpha$ with the choices above. Since $u^m = \mathbf{1}$, Equation (7.5) gives

$$\sum_{\ell=0}^{m-1} u^\ell s u^{-\ell} = \frac{1}{m} \sum_{k=0}^{m-1} u^k \left(\sum_{j=0}^{m-1} s_j \right) u^{-k} \leq \mathbf{1}. \quad (7.6)$$

Let $p_1, \dots, p_m$ be the spectral projections of u for the m th roots, allowing zero projections. Compressing Equation (7.6) by p_i gives $m p_i s p_i \leq p_i$, as required. $\square$

Corollary 7.3. *Under the hypotheses of Lemma 7.2, let P act normally on a Hilbert space and let $x = x^* \in P$ satisfy $\|x\| \leq 1$. For each $m \geq 2$, the operators sxs supplied by that lemma approximate x strongly, with associated m -partitions $\mathcal{P}$ satisfying*

$$\|sxs\| \leq 1, \quad \|C_{\mathcal{P}}(sxs)\| \leq 1/m.$$

Here the approximation is on any prescribed finite set of vectors, to any positive accuracy, while m stays fixed.

Proof. For test vectors $\xi_1, \dots, \xi_d$, normalize the sum of their vector functionals and those of $x\xi_1, \dots, x\xi_d$ to obtain a normal state θ . If this sum vanishes, all tests are zero and one may take $s = 0$. Otherwise Lemma 7.2 makes

$$\theta((1-s)^2) \leq 1 - \theta(s)$$

arbitrarily small. Thus $1 - s$ is arbitrarily small on every test vector and its image under x . The identity

$$sxs - x = sx(s-1) + (s-1)x$$

proves the strong approximation. Finally, $-s^2 \leq sxs \leq s^2$ and $s^2 \leq s$ imply $-p_i/m \leq p_i s x s p_i \leq p_i/m$. The norm of the block diagonal sum is the maximum of its block norms. $\square$

7.3 Assembly of the three corners

Lemma 7.4. *For any MASA $A \subset M$, set*

$$e = \bigvee \{\text{supp}(\lambda) : \lambda \in M_*^+, \lambda \text{ centralizes } A\}, \quad h = \mathbf{1} - e.$$

Then $e, h \in A$. If $e \neq 0$, there is a faithful normal conditional expectation $E : eMe \rightarrow Ae$. If $h \neq 0$, the inclusion $Ah \subset hMh$ has no nonzero centralizing normal positive functional.

Proof. The support of an A -central normal positive functional is invariant under every unitary in A , and hence belongs to A by maximal abelianness. Such a functional is faithful on its support corner, and remains faithful and centralizing upon restriction to any nonzero A -subprojection of its support.

Choose a maximal orthogonal family (q_β) of nonzero projections in A , each subordinate to the support of one such functional. Their sum is e : a nonzero remainder under e would have nonzero intersection with one of the defining supports, contradicting maximality. On each $q_\beta M q_\beta$, normalize the restricted functional to a faithful normal state. The corner $A q_\beta$ is a MASA in its centralizer. Takesaki's conditional expectation theorem [Takesaki72], in the state formulation [Hiai20, Theorem 6.6], gives a faithful normal expectation $E_\beta : q_\beta M q_\beta \rightarrow A q_\beta$. Its hypotheses hold because the centralizer is fixed by the modular group and a state is finite on the subalgebra.

Define

$$E(z) = \sum_{\beta} E_\beta(q_\beta z q_\beta), \quad z \in eMe. \quad (7.7)$$

The orthogonal block sum is well defined, bounded in norm by $\|z\|$, and belongs to Ae . It is positive, unital, Ae -bimodular, and fixes Ae . For a bounded increasing net of positive elements, normality of every E_β identifies the supremum of every q_β -block; this proves normality of E , even for an uncountable family. If $z \geq 0$ and $E(z) = 0$, faithfulness on each corner gives $q_\beta z q_\beta = 0$, hence $z^{1/2} q_\beta = 0$ for all β . Since $\sum_{\beta} q_\beta = e$, this forces $z = 0$. Thus E is faithful.

An Ah -central normal positive functional on hMh would extend by $z \mapsto hzh$ to an A -central one on M supported under h . The definition of e excludes a nonzero such functional. All corner MASAs used here remain maximal abelian: an element in a corner commuting with the corner of A , extended by zero, commutes with A . $\square$

Proof of Theorem 1.1. If $x = 0$, take $y = a = 0$ and the one-part partition, padded with zero projections to the size chosen below. Otherwise, by scaling it suffices to prove the assertion for $\|x\| = 1$, with an arbitrary strong-approximation tolerance $\delta > 0$. Fix $0 < \varepsilon < 1$ and put $\gamma = \varepsilon/4$. Let r_f be the partition-size bound $R_{\text{st}}(\gamma)$ in Theorem 6.1, and fix $m \geq 2$ with $1/m \leq \gamma$. We shall use

$$R(\varepsilon) = r_f + m + 1. \quad (7.8)$$

More explicitly, the choices in the state-paving proof may be taken as

$$b = \sigma = \frac{\varepsilon}{96(K+1)}, \quad r = \left\lceil \frac{32}{b^2} \right\rceil, \quad n = \left\lceil \frac{1024}{\varepsilon^2} \right\rceil, \quad m = \left\lceil \frac{4}{\varepsilon} \right\rceil.$$

Here $2\sqrt{2/r} + 2/r \leq b/2 + b^2/16 < b$, $4/\sqrt{n} \leq \gamma/2$, and $1/m \leq \gamma$. Consequently $r_f = (r+1)^2 n$ and

$$R(\varepsilon) = (r+1)^2 n + m + 1 \leq C_{\text{ap}} \varepsilon^{-6} \quad (0 < \varepsilon < 1) \quad (7.9)$$

for a universal constant C_{ap} . Only the universal buffer K enters this constant; graph, slab, and approximation parameters do not enter the color count. These choices precede the finite test set and its tolerance.

Let e, h and E be as in Lemma 7.4. Given a finite set F of test vectors, choose a unit vector ξ_0 such that $\|x\xi_0\| > 3/4$, and put $F' = F \cup \{\xi_0\}$. On Ae consider the normal positive functional

$$\psi(a) = \sum_{\xi \in F'} \langle ae\xi, e\xi \rangle, \quad a \in Ae,$$

and let $f = \text{supp}(\psi) \in Ae$ and $g = e - f$. The equality $\psi(g) = 0$ implies $g\xi = 0$ for every $\xi \in F'$. Empty corners below are interpreted as zero.

If $f \neq 0$, then $\psi|_{Af}$ is faithful. After normalizing it, compose it with the restriction $E : fMf \rightarrow Af$ to obtain a faithful normal state φ_f on fMf . Bimodularity of E and commutativity of Af show that Af is in the φ_f -centralizer. Apply Theorem 6.1 and Lemma 7.1 to the self-adjoint contraction $x_f = fxf$. They give

$$y_f = x_f - w, \quad w = w^*, \quad \|w\| \leq 2, \quad \|C_{\mathcal{P}_f}(y_f) - d_f\| \leq \gamma, \quad d_f = E(x_f),$$

where $\mathcal{P}_f$ is an r_f -partition of f , and w can be chosen arbitrarily small on the vectors $f\xi$ for $\xi \in F'$. Indeed, the bounded self-adjoint corrections in that theorem have state norm tending to zero, hence converge strongly in the given normal representation.

On g put $y_g = 0$. On h put $x_h = h x h$ and use Corollary 7.3 to choose $y_h = s x_h s$ arbitrarily close to x_h on the vectors $h\xi$, $\xi \in F'$, together with an m -partition $\mathcal{P}_h$ of h such that

$$\|y_h\| \leq 1, \quad \|C_{\mathcal{P}_h}(y_h)\| \leq \gamma.$$

Keep all off-diagonal blocks of x between f, g, h , and replace its diagonal blocks by the operators just constructed. Explicitly, set

$$y = y_f + y_g + y_h + \sum_{\substack{b, c \in \{f, g, h\} \\ b \neq c}} b x c.$$

This operator is self-adjoint. The difference $y - x$ is block diagonal: its f -block has norm at most 2, its g -block has norm at most 1, and its h -block has norm at most 2. Hence

$$\|y - x\| \leq 2, \quad \|y\| \leq 3. \quad (7.10)$$

The g -block of the difference annihilates F' . Choosing the other two approximation errors sufficiently small therefore ensures

$$\|(y - x)\xi\| < \min\{\delta, 1/4\} \quad (\xi \in F'). \quad (7.11)$$

In particular $\|y\| > 1/2$, by applying this inequality to ξ_0 .

Use the projections of $\mathcal{P}_f$, the projections of $\mathcal{P}_h$, and the single projection g as distinct members of one A -partition $\mathcal{P}$, padding with zeros in absent corners. It has the size in Equation (7.8) and kills every off-diagonal block between f, g, h . With $d = d_f$ on f and zero on $g + h$, we obtain

$$d = d^* \in A, \quad \|C_{\mathcal{P}}(y) - d\| \leq \gamma.$$

Since compression is norm-contractive, $\|d\| \leq \|y\| + \gamma$. Clip d by functional calculus to the interval $[-\|y\|, \|y\|]$, obtaining $a \in A$ with $\|a\| \leq \|y\|$ and $\|a - d\| \leq \gamma$. It follows that

$$\|C_{\mathcal{P}}(y) - a\| \leq 2\gamma = \varepsilon/2 \leq \varepsilon \|y\|.$$

Equations (7.10) and (7.11) give the remaining requirements. The partition size depends only on ε , since the bound in Theorem 6.1 is uniform over the state-preserving inclusions used here. Finally, scaling y and a by the original $\|x\|$, and first dividing the requested strong tolerance by that norm, proves the theorem with the universal constant $\kappa = 3$. $\square$

7.4 Consequences for related paving formulations

The main theorem concerns norm paving of bounded approximants. The two introductory consequences use the normal-expectation hypothesis to pass between that formulation and strong-operator paving of the original operator. The quadratic consequence additionally uses the separate companion theorem.

Proof of Corollary 1.2. By Theorem 1.1, every such x is $(\varepsilon/12, R(\varepsilon/12); 3)$ -approximately pable in the terminology of [PV15, Definition 2.3]. Apply [PV15, Proposition 2.4(1)] with $\varepsilon' = \varepsilon/6$. Its error is $2 \cdot 3 \cdot (\varepsilon/6) = \varepsilon$, so $R_s(\varepsilon) = R(\varepsilon/12)$ suffices. $\square$

Proof of Corollary 1.3. By [OpenAI26, Theorem 1.1], every self-adjoint $x \in M$ is $(\varepsilon/2, r_{\varepsilon/2})$ so-pable, where $r_{\varepsilon/2} \leq C\varepsilon^{-2}$ for a universal C . Since $\varepsilon/2 < \varepsilon$, [PV15, Proposition 2.4(2)] makes x $(\varepsilon, r_{\varepsilon/2}; 3)$ -approximately pable. Its proof supplies self-adjoint approximants and diagonal witnesses, with paving error relative to the approximant's norm. The strong-limit net in [PV15, Definition 2.3] supplies the finite-vector conclusion with the fixed count $N(\varepsilon) = r_{\varepsilon/2}$. $\square$

References

- [AW15] M. Aizenman and S. Warzel, *On the ubiquity of the Cauchy distribution in spectral problems*, Probab. Theory Related Fields **163** (2015), 61–87. doi:10.1007/s00440-014-0587-3; arXiv:1312.7769v2 (cited proposition numbering).
- [AS12] C. A. Akemann and D. Sherman, *Conditional expectations onto maximal abelian $*$ -subalgebras*, J. Operator Theory **68** (2012), no. 2, 597–607. Journal PDF.
- [Anderson79] J. Anderson, *Extensions, restrictions, and representations of states on C^* -algebras*, Trans. Amer. Math. Soc. **249** (1979), no. 2, 303–329. doi:10.2307/1998793.
- [Araki74] H. Araki, *Some properties of modular conjugation operator of von Neumann algebras and a non-commutative Radon–Nikodym theorem with a chain rule*, Pacific J. Math. **50** (1974), no. 2, 309–354. Journal PDF.
- [Boole1857] G. Boole, *On the comparison of transcendents, with certain applications to the theory of definite integrals*, Philosophical Transactions of the Royal Society of London **147** (1857), 745–803. doi:10.1098/rstl.1857.0037.
- [FM77] J. Feldman and C. C. Moore, *Ergodic equivalence relations, cohomology, and von Neumann algebras. II*, Trans. Amer. Math. Soc. **234** (1977), no. 2, 325–359. doi:10.1090/S0002-9947-1977-0578730-2.
- [Hiai20] F. Hiai, *Concise lectures on selected topics of von Neumann algebras*, 2020, arXiv:2004.02383v1.
- [HU16] C. Houdayer and Y. Ueda, *Asymptotic structure of free product von Neumann algebras*, Math. Proc. Cambridge Philos. Soc. **161** (2016), no. 3, 489–516. doi:10.1017/S0305004116000396; arXiv:1503.02460v3.
- [KS59] R. V. Kadison and I. M. Singer, *Extensions of pure states*, Amer. J. Math. **81** (1959), no. 2, 383–400. doi:10.2307/2372748.
- [KST99] A. S. Kechris, S. Solecki and S. Todorcevic, *Borel chromatic numbers*, Adv. Math. **141** (1999), no. 1, 1–44. doi:10.1006/aima.1998.1771.
- [Maharam64] D. Maharam, *Incompressible transformations*, Fund. Math. **56** (1964), no. 1, 35–50. doi:10.4064/fm-56-1-35-50.
- [MSS15] A. W. Marcus, D. A. Spielman and N. Srivastava, *Interlacing families II: Mixed characteristic polynomials and the Kadison–Singer problem*, Ann. of Math. (2) **182** (2015), no. 1, 327–350. doi:10.4007/annals.2015.182.1.8; arXiv:1306.3969v4.
- [OpenAI26] OpenAI, *Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras*, OpenAI Math Release preprint OAI:Quadratic-Strong-Operator-Paving-over-Arbitrary-Maximal-Abelian-Subalgebras-September-25-2026, 2026.
- [PSZ09] A. Poltoratski, B. Simon and M. Zinchenko, *The Hilbert transform of a measure*, 2009, arXiv:0811.0791v2.
- [Popa14] S. Popa, *A II_1 factor approach to the Kadison–Singer problem*, Comm. Math. Phys. **332** (2014), 379–414. doi:10.1007/s00220-014-2055-4; arXiv:1303.1424v3.

- [PV15] S. Popa and S. Vaes, *Paving over arbitrary MASAs in von Neumann algebras*, Anal. PDE **8** (2015), no. 4, 1001–1023. doi:[10.2140/apde.2015.8.1001](https://doi.org/10.2140/apde.2015.8.1001); arXiv:[1412.0631v3](https://arxiv.org/abs/1412.0631v3).
- [PV16] S. Popa and S. Vaes, *On the optimal paving over MASAs in von Neumann algebras*, Contemp. Math. **671** (2016), 199–208. doi:[10.1090/conm/671/13511](https://doi.org/10.1090/conm/671/13511); arXiv:[1507.01072v1](https://arxiv.org/abs/1507.01072v1).
- [RL20] M. Ravichandran and J. Leake, *Mixed determinants and the Kadison–Singer problem*, Math. Ann. **377** (2020), 511–541. doi:[10.1007/s00208-020-01986-7](https://doi.org/10.1007/s00208-020-01986-7); arXiv:[1609.04195v2](https://arxiv.org/abs/1609.04195v2) (cited theorem numbering).
- [RS21] M. Ravichandran and N. Srivastava, *Asymptotically optimal multi-paving*, Int. Math. Res. Not. IMRN **2021**, no. 14, 10908–10940. doi:[10.1093/imrn/rnz111](https://doi.org/10.1093/imrn/rnz111); arXiv:[1706.03737v2](https://arxiv.org/abs/1706.03737v2) (cited theorem numbering).
- [RS06] P. Roy and G. Samorodnitsky, *Maharam extension for nonsingular group actions*, 2006, arXiv:[math/0611891v1](https://arxiv.org/abs/math/0611891v1).
- [Takesaki72] M. Takesaki, *Conditional expectations in von Neumann algebras*, J. Funct. Anal. **9** (1972), no. 3, 306–321. doi:[10.1016/0022-1236\(72\)90004-3](https://doi.org/10.1016/0022-1236(72)90004-3).